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Facet sensitivity of iron carbides in Fischer-Tropsch synthesis

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Fischer-Tropsch synthesis (FTS) is a structure-sensitive reaction of which performance is strongly related to the active phase, particle size, and exposed facets. Compared with the full-pledged investigation on the active phase and particle size, the facet effect has been limited to theoretical studies or single-crystal surfaces, lacking experimental reports of practical catalysts, especially for Fe-based catalysts. Herein, we demonstrate the facet sensitivity of iron carbides in FTS. As the prerequisite, {202} and {112} facets of χ-Fe_5C_2 are fabricated as the outer shell through the conformal reconstruction of Fe_3O_4 nanocubes and octahedra, as the inner cores, respectively. During FTS, the activity and stability are highly sensitive to the exposed facet of iron carbides, whereas the facet sensitivity is not prominent for the chain growth. According to mechanistic studies, {202} χ-Fe_5C_2 surfaces follow hydrogen-assisted CO dissociation which lowers the activation energy compared with the direct CO dissociation over {112} surfaces, affording the high FTS activity.

Fischer-Tropsch synthesis (FTS) is a structure-sensitive reaction for the sustainable production of synthetic fuels and building-block chemicals from syngas[1–6]. The catalytic performance is strongly related to the active phase, particle size, and exposed facets of the active components such as iron, cobalt, and ruthenium[7–13]. Compared with Co and Ru, Fe-based catalysts have superior properties including resistance to the formation of methane, low cost, high adaptability to broad H_2/CO ratios, and versatility to various useful products[14–16]. For Fe-based catalysts, pure-phase iron carbides were synthesized by using $Fe(CO)_5$ reagent, which was explored by means of in-situ characterizations[17,18]. The Fe_3O_4@χ-Fe_5C_2 core-shell catalysts were constructed by pyrolyzing iron-containing metal-organic frameworks, whereas the obtained nanoparticles were irregular spherical particles[19]. The transformation of reduced iron phases to iron carbides promoted the formation of hydrocarbon species in FTS[20]. The effects of the active phase and particle size have been extensively studied[19,21–26]. These effects are generally entangled with the contributions of surface terrace, corner, edge, and step-edge sites, where differences in coordination numbers and surface topology may lead to substantial differences in intrinsic performance[27]. Up to date, the investigation of the facet effect has been limited to theoretical calculations or single-crystal surfaces. For instance, density functional theory (DFT) calculations of CO activation on χ-Fe_5C_2 surfaces indicated that the terraced (510) surface inclined to directly dissociate CO molecules, whereas the stepped (010) and (001) surfaces preferred the hydrogen-assisted CO dissociation route[28]. In-situ scanning tunneling microscopy visualized on-surface ethylene polymerization on a carburized Fe(110) single-crystal surface[29]. However, there is no experimental report on the facet effect of practical Fe-based catalysts due to the complexity and dynamic structural evolution of iron carbides during FTS.

As the widely accepted active phase for FTS, χ-Fe_5C_2 has a base-centered monoclinic (*bcm*) structure with space group C2/*c* ($a = 11.56$ Å, $b = 4.57$ Å, $c = 5.06$ Å, and $\beta = 97.74°$)[30,31]. Owing to the low symmetry of the lattice structure, it remains as a grand challenge to

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synthesize χ-Fe_5C_2 nanocrystals with uniformly exposed surfaces. We proposed to use a highly symmetrical template as the core to support the χ-Fe_5C_2. Fe_3O_4 has a face-centered cubic (*fcc*) structure with a high symmetry. Herein, we reported the conformal reconstruction of well-defined Fe_3O_4 nanocrystals to generate χ-Fe_5C_2 with specifically exposed surfaces. The samples consisted of an inner core of Fe_3O_4 and an outer shell of χ-Fe_5C_2, denoted as Fe_3O_4@χ-Fe_5C_2 nanocrystals. We obtained Fe_3O_4@χ-Fe_5C_2 nanocrystals with surfaces terminated in {202} and {112} facets of χ-Fe_5C_2 shells through using cubic and octahedral Fe_3O_4 as the templates, respectively (Fe_3O_4@χ-Fe_5C_2 nanocubes and octahedra, respectively). We discovered that Fe_3O_4@χ-Fe_5C_2 nanocubes were more catalytically active and stable than the octahedral counterpart during FTS, whereas the facet sensitivity was not prominent for the chain growth. According to mechanistic studies, the high activity of {202} χ-Fe_5C_2 surfaces derived from the unique reaction path in which the hydrogen-assisted CO dissociation route lowered the activation energy relative to the direct CO dissociation route over {112} surfaces.

Results and discussion

Synthesis and characterization of Fe_3O_4@χ-Fe_5C_2 nanocubes

To begin with, Fe_3O_4 nanocubes were prepared with an average size of 40.5 ± 3.9 nm and a purity of 95.3% (Supplementary Fig. 1). Fe_3O_4@χ-Fe_5C_2 core-shell nanocubes were synthesized via surface reconstruction of Fe_3O_4 under syngas atmosphere. Specifically, Fe_3O_4 nanocubes were reduced under 1 bar of H_2 with a gas-flow rate of 100 mL min^{-1} at 270 °C for 10 h. This treatment ensured the removal of surface organic species (Supplementary Fig. 2). Afterwards, the obtained samples underwent surface reconstruction in a fixed-bed reactor under 20 bar of syngas (32 vol% H_2, 64 vol% CO, and 4 vol% Ar) with a space velocity of 2400 mL h^{-1} g_{cat}^{-1} at 270 °C for 20 h. Figure 1a shows the transmission electron microscopy (TEM) image of Fe_3O_4@χ-Fe_5C_2 nanocubes.

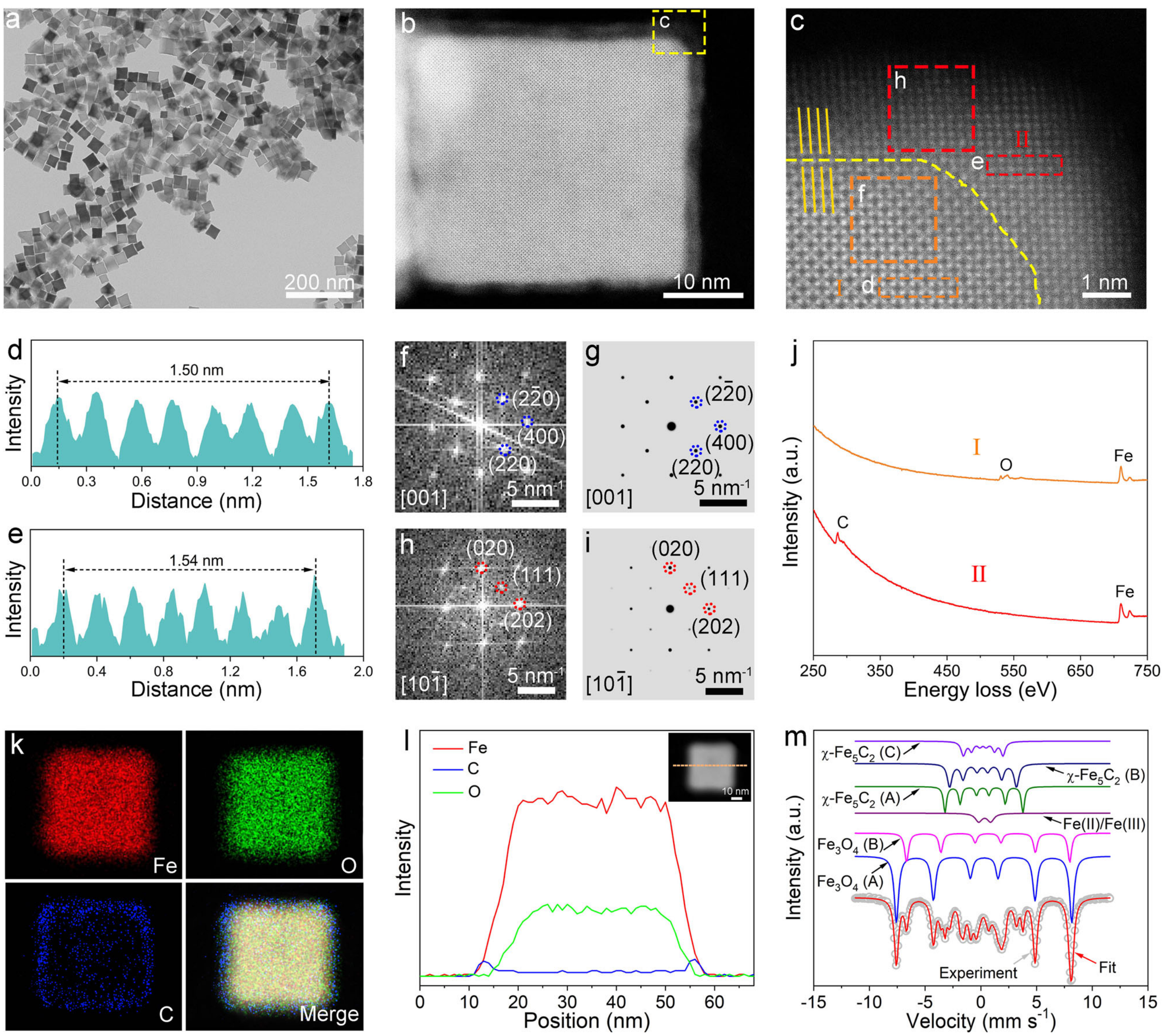

Fig. 1 | Structural characterizations of Fe_3O_4@χ-Fe_5C_2 nanocubes. **a** TEM image of Fe_3O_4@χ-Fe_5C_2 nanocubes. **b** HAADF-STEM image of an individual Fe_3O_4@χ-Fe_5C_2 nanocube. **c** Magnified HAADF-STEM image of the region marked by the corresponding box in panel (**b**). **d** Intensity profile recorded from the area indicated by the rectangular box in panel (**c**). **e** Intensity profile recorded from the area indicated by the rectangular box in panel (**c**). **f** FFT pattern from box **f** in panel (**c**). **g** Simulated FFT pattern of Fe_3O_4 along the [001] direction. **h** FFT pattern from box **h** in panel (**c**). **i** Simulated FFT pattern of χ-Fe_5C_2 along the [10-1] direction. **j** EELS spectra of a Fe_3O_4@χ-Fe_5C_2 nanocube in panel (**c**). **k** Elemental mapping images of a Fe_3O_4@χ-Fe_5C_2 nanocube. **l** Line-scan profile recorded along the line of the inset HAADF image. **m** Mössbauer spectra of Fe_3O_4@χ-Fe_5C_2 nanocubes.

The average size of Fe_3O_4@χ-Fe_5C_2 nanocubes was 40.1 ± 3.8 nm with a purity of 90.0% (Supplementary Fig. 3). The Fe_3O_4@χ-Fe_5C_2 nanocubes exhibited a surface area of 20.4 $m^2\,g^{-1}$ based on the Brunauer-Emmett-Teller (BET) method (Supplementary Fig. 4a). The high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) image taken from one of the nanocubes revealed a periodic lattice extending across the entire surface (Fig. 1b). The magnified image clearly revealed lattice differences between the edge region and the central region, implying the formation of the core-shell structure (Fig. 1c). In the case of nanocrystals with a core-shell structure, a slight lattice mismatch (f < ~5%) is required for the epitaxial surface layer to form over the inner core[32]. This epitaxial relationship allows for the maintenance of orientation between the growth layer and the substrate within the first few atomic layers. The spacing of Fe_3O_4(400) planes (0.21 nm) in the core was approximately equal to that of χ-Fe_5C_2(202) planes (0.22 nm) in the shell (Fig. 1d, e). The small lattice mismatch (f) of 4.65% calculated from Eq. (1) ensures the preservation of the epitaxial orientation relationship.

$$f = 2 \times |d_{\mathrm{shell}} - d_{\mathrm{core}}|/(d_{\mathrm{shell}} + d_{\mathrm{core}}) \tag{1}$$

In Eq. (1), d_{shell} and d_{core} refer to the lattice spacings of the shell and the core, respectively. The small lattice mismatch results in only one type of facet at the surface layer of Fe_3O_4@χ-Fe_5C_2 nanocubes. To identify the corresponding facets, we conducted the fast Fourier transform (FFT) analysis. According to the combination of the experimental and simulated FFT patterns, the region in the inner core was indexed as the (400) facet of *fcc* Fe_3O_4 along the [001] zone axis, while that in the outer shell corresponded to the (202) facet of *bcm* χ-Fe_5C_2 along the [10−1] zone axis (Fig. 1f–i). The HAADF-STEM images of different corners and edges in an individual nanoparticle indicated that the surfaces exposed a uniform χ-Fe_5C_2 shell with the {202} facets (Supplementary Fig. 5). The average thickness of the shell was determined as 2.0 nm, approximating ten atomic layers (Supplementary Fig. 5).

The core of iron oxides and the shell of iron carbides were further confirmed through the spatial elemental analysis. Specifically, the electron energy loss spectroscopy (EELS) image implied that the core region mainly comprised Fe and O elements while the shell region contained Fe and C elements (Fig. 1j). The scanning transmission electron microscopy-energy dispersive X-ray (STEM-EDX) analysis including the elemental mapping images and the line-scanning profile confirmed that O and C elements were mainly located in the core and the shell, respectively, while Fe was homogeneously distributed throughout the particle (Fig. 1k, l).

To quantify the contents of different phases in Fe_3O_4@χ-Fe_5C_2 nanocubes, we carried out Mössbauer and X-ray diffraction (XRD) characterizations. Based on the Mössbauer results, the sample was composed of 63.4 wt% of Fe_3O_4, 33.2 wt% of χ-Fe_5C_2, and 3.4 wt% of Fe(II)/Fe(III) species (Fig. 1m and Supplementary Table 1). Fe(II)/Fe(III) species indicated the presence of some poorly crystallized iron oxides. The quantitative analysis of XRD showed that χ-Fe_5C_2 occupied 29.8 wt% of the total mass, approaching that (33.2 wt%) obtained from the Mössbauer spectra (Supplementary Fig. 6a and Supplementary Table 2).

Evolution from Fe_3O_4 nanocubes to Fe_3O_4@χ-Fe_5C_2 nanocubes

Figure 2a depicts the schematic of the evolution from Fe_3O_4 nanocubes to Fe_3O_4@χ-Fe_5C_2 nanocubes. The major steps were verified by HAADF-STEM images and XRD patterns. Specifically, after reduction, the initial Fe_3O_4 nanocubes with {400} facets were transformed into metallic Fe nanocubes with their surfaces terminated in {100} facets (Fig. 2b, c and Supplementary Fig. 6b, c). When metallic Fe nanocubes were exposed to the syngas, the thermodynamic driving force induced the phase transition from metallic Fe to iron carbides. The surface Fe atoms underwent carburization and oxidation by reacting with carbon and oxygen species derived from the dissociated CO. After the exposure to syngas for 2 h, the corners were preferentially carburized and oxidized, resulting in the random distribution of Fe_3O_4 and χ-Fe_5C_2 domains (Fig. 2d). Notably, the diffractions of Fe_3O_4 and metallic Fe were clearly observed, whereas the crystalline χ-Fe_5C_2 was not detected by XRD (Supplementary Fig. 6d). This result implied that χ-Fe_5C_2 existed in short-range order with an amorphous structure. When Fe nanocubes were exposed to syngas for 5 h, carbon atoms cleaved from CO completely carburized the metallic Fe at the corner of the nanocube, while the dissociated oxygen atoms permeated into the interior, resulting in the bulk transition into Fe_3O_4 (Fig. 2e). At this stage, the diffractions of χ-Fe_5C_2 were observed in the XRD patterns, along with metallic Fe and Fe_3O_4 (Supplementary Fig. 6e). When Fe nanocubes were exposed to syngas for 10 h, the dissociated carbon atoms carburized the surface layers from the corners to the whole faces (Fig. 2f and Supplementary Fig. 6f). A core-shell structure with the Fe_3O_4 core and χ-Fe_5C_2 surface formed as the result of that the nanocubes lost the thermodynamic driving force for further carburization or oxidation[33]. The core-shell structure did not show obvious change when the syngas treatment was prolonged to 20 h (Fig. 1b, c). When the Fe_3O_4@χ-Fe_5C_2 core-shell structure formed at a steady state, the excess dissociated oxygen atoms were released into the gas phase in the form of CO_2 and H_2O to prevent the oxidation of the shell. The excess dissociated carbon atoms reacted with surface-dissociated hydrogen atoms to yield hydrocarbons rather than to permeate and carburize the Fe_3O_4 interlayer. The stable core-shell structure was the result of the dynamic balance of the hydrocarbon production, surface oxidation, and carburization in the syngas environment.

We conclude the driving force to limit a single type of phase and facet at the surface layer as follows. The driving force to regulate the phase is the carbon chemical potential of reaction conditions. It was reported that the activation energy for carbon diffusion in Fe (43.9–69.0 kJ mol^{-1}) was lower than that for the FTS reaction (89.1 ± 3.8 kJ mol^{-1})[14]. Consequently, surface carbon atoms cleaved from CO exhibit a pronounced affinity to Fe atoms, thereby instigating a phase transition from iron to FeC_x during the initial stage of the FTS reaction. Upon the formation of active FeC_x on the surface, the FTS reaction is facilitated, resulting in the release of oxygen species, predominantly in the form of H_2O, into the gas phase. This influx of H_2O induces Fe oxide formation and impedes further carburization. As the concentration of oxygen species diminishes in the gas phase, a greater quantity of carbide accumulates on the surface, pushing the reaction condition back to the equilibrium position and vice versa. As the inner core of the catalyst oxidizes to Fe_3O_4, it becomes more resistant to carbon permeation and carburization compared to metallic Fe. According to carbon chemical potential theory, carbon-rich χ-Fe_5C_2 is the thermodynamically stable phase under our reaction condition (20 bar, CO:H_2 = 1:2, 270 °C)[21]. As such, other Fe carbides present in the surface layers will evolve into χ-Fe_5C_2 as the reaction progresses. Moreover, the uniform facet of the substrate and the small lattice mismatch (<~5%) within the core-shell structure ensure the preservation of the epitaxial orientation relationship.

Synthesis and characterization of Fe_3O_4@χ-Fe_5C_2 octahedra

For comparison, we applied the surface reconstruction procedure to prepare Fe_3O_4@χ-Fe_5C_2 octahedra. We carbonized the synthesized Fe_3O_4 octahedra following a similar approach to that of nanocubes (Supplementary Fig. 7). The average size of Fe_3O_4@χ-Fe_5C_2 octahedra was 45.4 ± 3.5 nm with a purity of 93.3% (Fig. 3a and Supplementary Fig. 8). The surface area of Fe_3O_4@χ-Fe_5C_2 octahedra was 22.0 $m^2\,g^{-1}$ based on the BET method (Supplementary Fig. 4b). The lattice disparity between the core and the shell was clearly revealed by the HAADF-STEM images (Fig. 3b and Supplementary Fig. 9). The uniform

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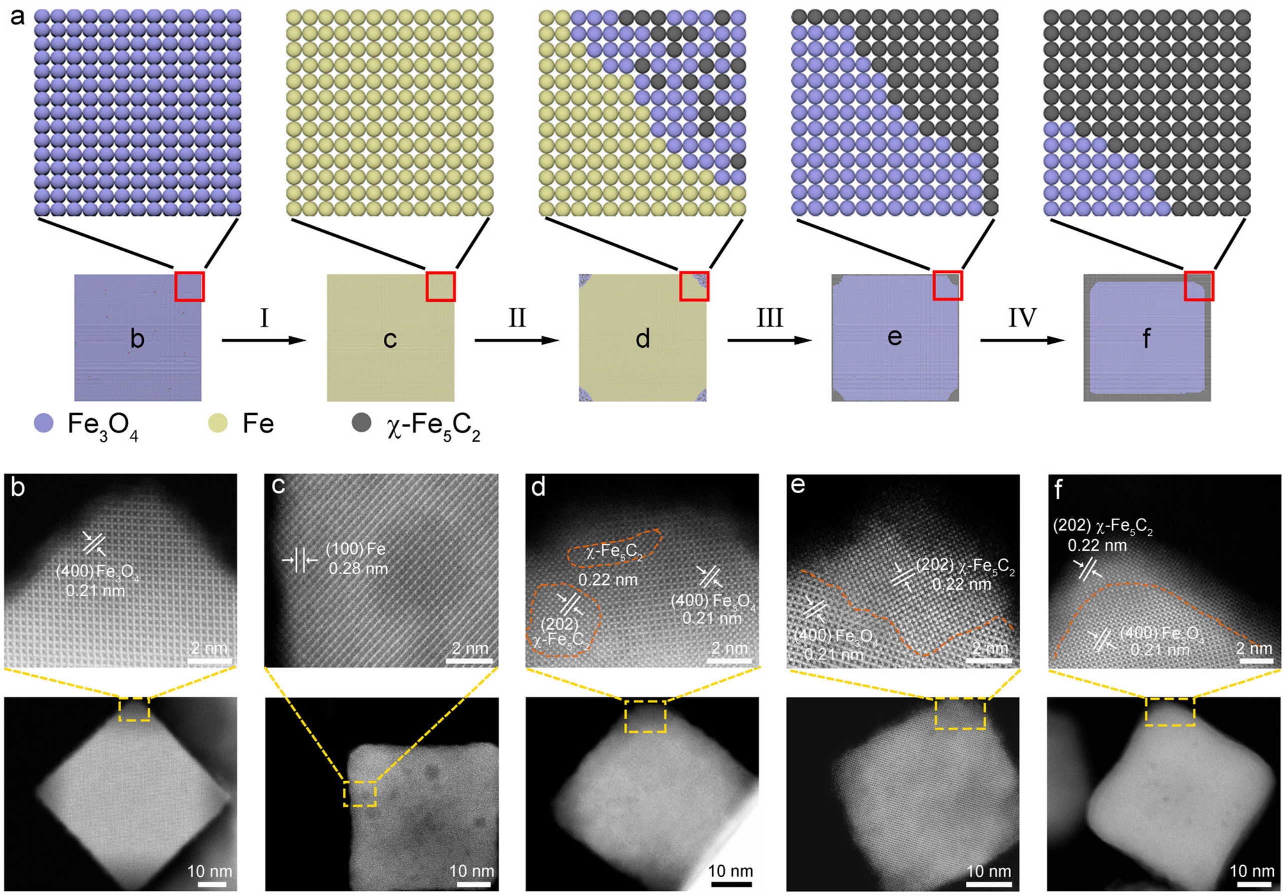

Fig. 2 | Structural characterizations of the evolution from the Fe_3O_4 nanocube to the Fe_3O_4@χ-Fe_5C_2 nanocube. a Schematic of the major steps involved in the continuous evolution from the Fe_3O_4 nanocube to the Fe_3O_4@χ-Fe_5C_2 nanocube. I refers to the reduction process; II refers to the corner carburization and oxidation; III refers to the surface carburization, O migration, and bulk oxidation; IV refers to the surface balanced carburization and oxidation. **b** HAADF-STEM images of an individual Fe_3O_4 nanocube. **c** HAADF-STEM images of an individual Fe nanocube. **d**–**f** HAADF-STEM images of an individual nanocrystal after the treatment of Fe nanocubes with syngas for 2, 5, and 10 h, respectively.

shell had an average thickness of 1.7 nm, corresponding to eight atomic layers (Fig. 3c). We assigned the lattice parameters with the help of the experimental and simulated FFT patterns. Specifically, the inner core took a lattice parameter of 0.48 nm which was indexed as the (111) facet of *fcc* Fe_3O_4 along the [001] zone axis (Fig. 3d, f, g). For the outer shell, the lattice parameter of 0.21 nm was assigned to the (112) facet of *bcm* χ-Fe_5C_2 along the [10-1] zone axis (Fig. 3e, h, i). Besides lattice matching, the epitaxial orientation can also be preserved via domain matching, where the spacing of *m* lattice planes in the epilayer is approximately equal to *n* in the substrate[32,34,35]. We observed that the spacing of three Fe_3O_4(111) planes in the core was approximately equal to that of seven χ-Fe_5C_2(112) planes in the shell (Fig. 3c). Such periodicity leads to a commensurate epitaxial relationship with a low mismatch value of 2.06% according to Eq. (2).

$$f = 2 \times |7 \times d_{\mathrm{shell}} - 3 \times d_{\mathrm{core}}| / (7 \times d_{\mathrm{shell}} + 3 \times d_{\mathrm{core}}) \quad (2)$$

The domain match allows the conformal growth for the (112) facet of χ-Fe_5C_2 over the (111) facet of Fe_3O_4. The distribution of Fe_3O_4 at the core and χ-Fe_5C_2 at the shell was supported by the EELS image, STEM-EDX elemental mapping images, and line scanning profile (Fig. 3j–l). Mössbauer result indicated that Fe_3O_4@χ-Fe_5C_2 octahedra contained 65.1 wt% of Fe_3O_4 and 29.5 wt% of χ-Fe_5C_2 (Fig. 3m). The content of χ-Fe_5C_2 was consistent with the XRD quantitative analysis result (27.6%) (Supplementary Tables 1 and 2).

Facet effect on activity and selectivity

We explored the facet effect of iron carbides on FTS properties. As the shells of Fe_3O_4@χ-Fe_5C_2 nanocrystals are more than six atomic layers thick, the electronic coupling between the core and outermost layer in the shell is essentially lost, and thus the ability to access the strain-dependent catalytic activity will be gone[36]. Besides, it was worth noting that no promoters or additives were added since the purpose of this work was to investigate the intrinsic catalytic performance of different exposed facets of χ-Fe_5C_2. Fe_3O_4@χ-Fe_5C_2 nanocubes and octahedra were loaded on the SiC support, denoted as Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC and octahedra/SiC, respectively. The TEM images of these nanocrystals and the corresponding particle models from different orientations were shown in Supplementary Figs. 10 and 11. The catalytic properties of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC and octahedra/SiC were evaluated in a fixed-bed reactor under 20 bar of syngas (64 vol% H_2, 32 vol% CO, and 4 vol% Ar) with a space velocity of 2400 mL h^{-1} g_{cat}^{-1} at 270 °C, denoted as the standard condition. The CO conversion of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was 45.4%, which was higher than that (21.2%) of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC (Fig. 4a and Supplementary Table 3).

To compare the catalytic activity more accurately, we calculated the TOF numbers based on the moles of CO converted per mole of surface Fe atoms per hour. The moles of Fe atoms on the surface of Fe_3O_4@χ-Fe_5C_2 nanocrystals were determined by CO pulse chemisorption measurement. The moles of Fe atoms on the surface of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was 24.1 μmol g^{-1}, higher than that

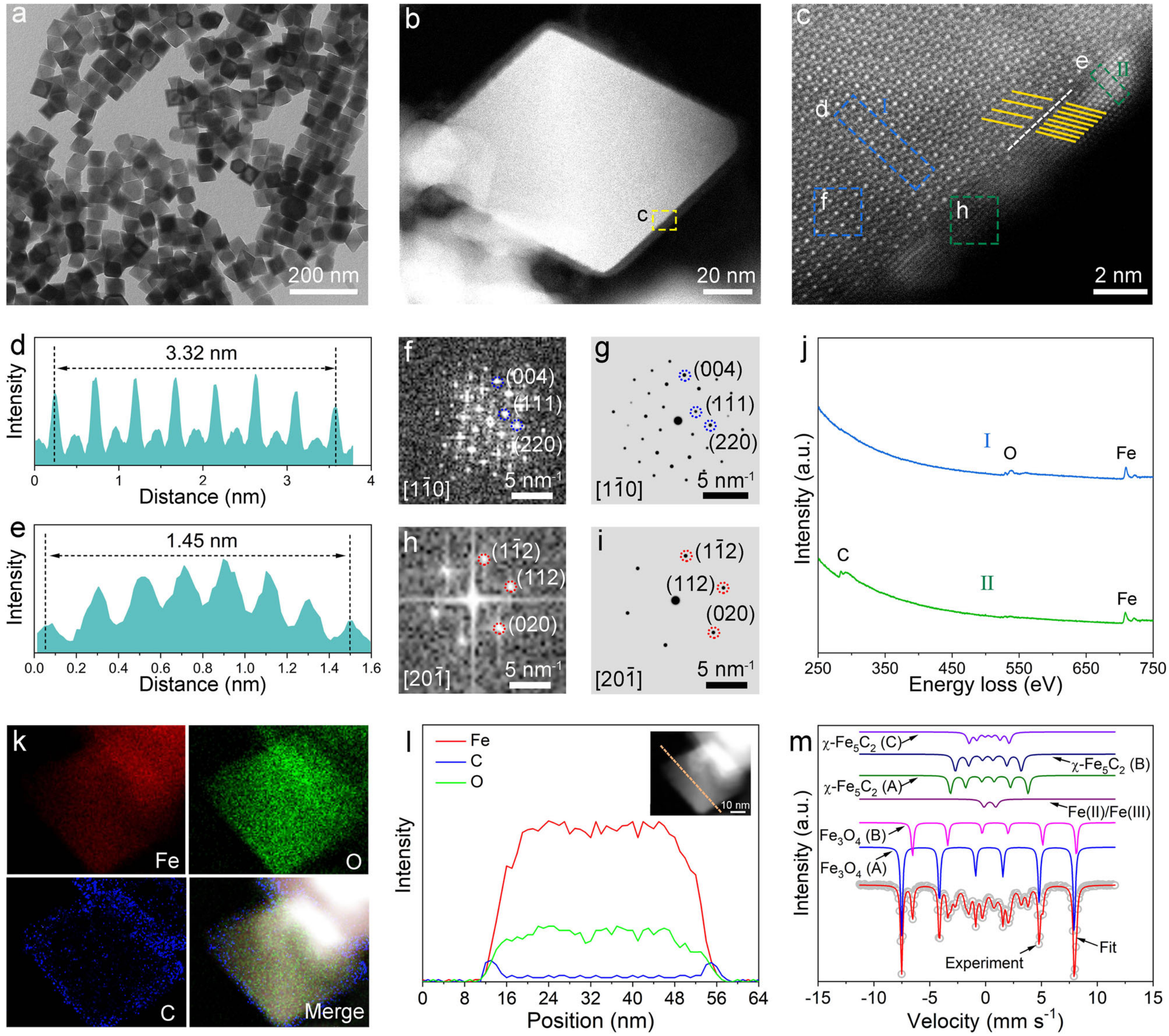

Fig. 3 | Structural characterizations of Fe_3O_4@χ-Fe_5C_2 octahedra. a TEM image of Fe_3O_4@χ-Fe_5C_2 octahedra. **b** HAADF-STEM image of an individual Fe_3O_4@χ-Fe_5C_2 octahedron. **c** Magnified HAADF-STEM image of the region marked by the corresponding box in panel (**b**). **d** Intensity profile recorded from the area indicated by the rectangular box in panel (**c**). **e** Intensity profile recorded from the area indicated by the rectangular box in panel (**c**). **f** FFT pattern from box **f** in panel (**c**). **g** Simulated FFT pattern from of Fe_3O_4 along the [1–10] direction. **h** FFT pattern from box **h** in panel (**c**). **g** Simulated FFT pattern from of χ-Fe_5C_2 along the [20–1] direction. **j** EELS spectra of a Fe_3O_4@χ-Fe_5C_2 octahedron in panel (**c**). **k** elemental mapping images of a Fe_3O_4@χ-Fe_5C_2 octahedron. **l** Line-scan profile recorded along the line of the inset HAADF image. **m** Mössbauer spectra of Fe_3O_4@χ-Fe_5C_2 octahedra.

(19.5 μmol g^{-1}) of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC (Supplementary Fig. 12a, b). The TOF number of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was 645.9 h^{-1}, being 1.7 times as high as that (372.7 h^{-1}) of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC. The carbon balance value of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was 98.7%, similar to that (96.5%) of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC (Supplementary Table 3). The high carbon balance value indicated that the different CO conversions were not caused by continuous carburization. For reference, we prepared pure χ-Fe_5C_2 nanoparticles which mainly exposed the thermodynamically most stable {510} facet through the wet-chemical route[30] (Supplementary Fig. 13). The CO conversion of χ-Fe_5C_2 nanoparticles/SiC was 23.6% (Fig. 4a and Supplementary Table 3). As such, the {202} χ-Fe_5C_2 facet exposed on Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was more active than the {112} χ-Fe_5C_2 facet exposed on Fe_3O_4@χ-Fe_5C_2 octahedra/SiC and thermodynamically stable surfaces of iron carbides.

With respect to the selectivity, the C_{5+} selectivity of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was 44.8 C%, higher than that (27.6 C%) of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC (Fig. 4a and Supplementary Table 3). The C_2-C_4 selectivity of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was 41.0 C%, lower than that (53.1 C%) of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC (Fig. 4a and Supplementary Table 3). Additionally, the ratio of olefins to paraffins (o/p ratio) among C_2-C_4 for Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was 1.1, higher than that (0.6) for Fe_3O_4@χ-Fe_5C_2 octahedra/SiC (Fig. 4a and Supplementary Table 3). The selectivity for C_2-$C_4^=$ olefins over Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was 21.6 C%, approaching that (20.6 C%) over Fe_3O_4@χ-Fe_5C_2 octahedra/SiC (Supplementary Table 3). Additionally, the selectivity for C_5–$C_{12}^=$ olefins over Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was 17.9 C%, higher than that (11.3 C%) over Fe_3O_4@χ-Fe_5C_2 octahedra/SiC (Supplementary Table 3). As for χ-Fe_5C_2 nanoparticles/SiC, the selectivities for CH_4, C_2-C_4, C_5-C_{12}, and C_{13+} were 15.5 C%, 48.2 C%, 34.9 C%, and 1.4 C%, respectively (Fig. 4a and Supplementary Table 3). The C_2-$C_4^=$ and C_5-$C_{12}^=$ selectivities were 29.8 C% and 16.9 C%, respectively (Supplementary Table 3). Actually, the distribution of hydrocarbon products for Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC,

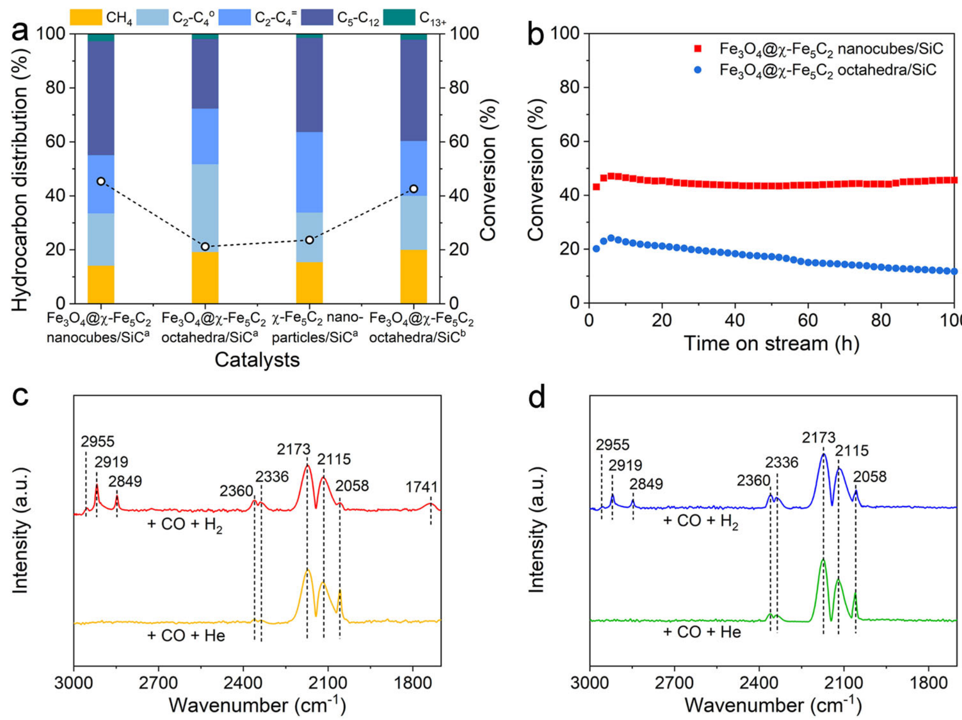

Fig. 4 | Catalytic properties and structural characterizations of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC and octahedra/SiC. a CO conversion and selectivity of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC, Fe_3O_4@χ-Fe_5C_2 octahedra/SiC, and χ-Fe_5C_2/SiC. [a] refers to that the reaction was conducted under 20 bar of syngas (CO:H_2 = 1:2, 2400 mL h^{-1} g_{cat}^{-1}) at 270 °C. [b] refers to that the reaction was conducted under 20 bar of syngas (CO:H_2 = 1:2, 800 mL h^{-1} g_{cat}^{-1}) at 270 °C. **b** Stability tests of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC and octahedra/SiC. The reaction was conducted under 20 bar of syngas (CO:H_2 = 1:2, 2400 mL h^{-1} g_{cat}^{-1}) at 270 °C. **c** In-situ DRIFTS spectra of Fe_3O_4@χ-Fe_5C_2 nanocubes and **d** Fe_3O_4@χ-Fe_5C_2 octahedra after being exposed to CO for 30 min and purged with He or H_2 for 30 min at 270 °C.

Fe_3O_4@χ-Fe_5C_2 octahedra/SiC, and χ-Fe_5C_2 nanoparticles/SiC followed a typical Anderson-Schulz-Flory (ASF) statistics. The probabilities of chain growth (α) for Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC, Fe_3O_4@χ-Fe_5C_2 octahedra/SiC, and χ-Fe_5C_2 nanoparticles/SiC were calculated as 0.66, 0.62, and 0.63, respectively (Supplementary Figs. 14, 15). To compare the catalytic selectivity more appropriately, we adjusted the space velocity of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC to keep the reaction over cubic and octahedral nanocrystals at similar conversion levels. When the space velocity was lowered to 800 mL h^{-1} g_{cat}^{-1}, CO conversion of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC reached 42.6%, approaching that (45.4%) of the cubic counterpart under the standard condition (Fig. 4a and Supplementary Table 3). The C_{5+} selectivity of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC was 39.6 C%, while the C_2-C_4 hydrocarbons occupied 40.3 C% of all hydrocarbons (Fig. 4a and Supplementary Table 3). The selectivities for C_2-$C_4^=$ and C_5-$C_{12}^=$ were 18.3 C% and 16.0 C%, respectively (Supplementary Table 3) The α value was calculated as 0.64 at this conversion level (Supplementary Fig. 16). Therefore, the effect of crystal face on selectivity is not as great as that on activity.

We evaluated the long-term stability of Fe_3O_4@χ-Fe_5C_2 nanocrystals/SiC through a 100-h test. The CO conversion of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC fluctuated within 1% during the whole test, indicating the high stability of this catalyst (Fig. 4b). The cubic morphology was preserved after 100 h on stream (Supplementary Fig. 17a, b). We measured the thickness of the shell layer for Fe_3O_4@χ-Fe_5C_2 nanocubes after the reaction. The thickness of the shell increased from 2.0 nm to 2.6 nm after the reaction (Fig. 1c and Supplementary Fig. 17c). The lattice parameter of the Fe_3O_4 inner core was measured as 0.21 nm, which was indexed as the (400) facet of Fe_3O_4 (Supplementary Fig. 17d). The lattice parameter of the χ-Fe_5C shell was 0.22 nm, which was assigned to the (202) facet of χ-Fe_5C_2 (Supplementary Fig. 17e). The exposed χ-Fe_5C_2 facets of Fe_3O_4@χ-Fe_5C_2 nanocubes were preserved after 100 h on stream. The EELS image implied that the core region mainly comprised Fe and O elements while the shell region contained Fe and C elements (Supplementary Fig. 17f). We also conducted Mössbauer spectroscopy to characterize the compositions of the iron phase in the used Fe_3O_4@χ-Fe_5C_2 nanocubes after 100 h on stream (Supplementary Fig. 18a). The content of χ-Fe_5C_2 increased from 33.2% to 39.8% after the reaction (Supplementary Table 4).

In contrast, the CO conversion of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC declined continuously. The conversion was only 11.7% after 100 h, about half of that (20.1%) at the beginning (Fig. 4b). To investigate the reason for the deactivation, we characterized Fe_3O_4@χ-Fe_5C_2 octahedra/SiC after 100 h on stream. As shown in Supplementary Fig. 19a, b, the solid octahedra collapsed after the reaction. The formation of Fe_3O_4@χ-Fe_5C_2 octahedra with multiple voids was ascribed to the Kirkendall effect[37–39]. Specifically, as the blockage of active sites by long-chain hydrocarbons and non-graphitic carbon, Fe_3O_4@χ-Fe_5C_2 octahedra were gradually deactivated. Meanwhile, the carbon chemical potential of reaction conditions changed. The dynamic balance of the hydrocarbon production, surface oxidation, and carburization in the syngas environment was broken. This led to the diffusion of Fe atoms between the Fe_3O_4 core and χ-Fe_5C_2 shell. The void formation in Fe_3O_4@χ-Fe_5C_2 octahedra due to differential diffusion rates of Fe atoms. The thickness of the shell increased from 1.7 nm to 3.5 nm after the reaction (Fig. 3c and Supplementary Fig. 19c). The lattice parameters of the inner core and the outer shell were measured as 0.48 nm and 0.21 nm, which were assigned to the (111) facet of Fe_3O_4 and (112) facet of χ-Fe_5C_2, respectively (Supplementary Fig. 19d, e). The exposed χ-Fe_5C facet of Fe_3O_4@χ-Fe_5C_2 octahedra was preserved after the reaction. The distribution of Fe_3O_4 at the core and χ-Fe_5C_2 at the shell was supported by the EELS image (Supplementary Fig. 19f). The content of χ-Fe_5C_2 in Fe_3O_4@χ-Fe_5C_2 octahedra increased from 29.5% to 40.4% after 100 h on stream (Supplementary Fig. 18b and Supplementary Table 4). The thickness of the shell layer for both Fe_3O_4@χ-Fe_5C_2 nanocubes and octahedra increased after the reaction.

Therefore, the stabilities of Fe_3O_4@χ-Fe_5C_2 nanocrystals/SiC were also affected by the exposed facets.

To investigate the textural properties, we carried out N_2 physisorption characterizations. The pore-diameter distributions were analyzed using the Barrett-Joyner-Halenda (BJH) method. As shown in (Supplementary Fig. 4c, d), the surface layer of both Fe_3O_4@χ-Fe_5C_2 nanocubes and octahedra were not porous. The absence of pores at the surface layer was also confirmed by HAADF-STEM images (Figs. 1b and 3b). We also measured the textural properties of spent Fe_3O_4@χ-Fe_5C_2 nanocubes and octahedra after 100 h on stream (Supplementary Fig. 20a, b). The surface layer of spent Fe_3O_4@χ-Fe_5C_2 nanocubes remained nonporous after 100 h on stream (Supplementary Fig. 20c). In contrast, spent Fe_3O_4@χ-Fe_5C_2 octahedra contained mesopores as revealed by the pore-diameter distribution and HAADF-STEM image (Supplementary Fig. 20d). The average mesopore diameter was determined as 16.0 nm by the BJH method (Supplementary Fig. 20d).

To investigate other underlying mechanisms for catalyst deactivation, we employed Raman spectroscopy to analyze the surfaces of Fe_3O_4@χ-Fe_5C_2 nanocubes and octahedra after 100 h. The presence of peaks at 1330 cm^{-1} indicated the existence of disordered carbon (D band), while those at 1592 cm^{-1} signified graphite (G band) (Supplementary Fig. 21a, b). Significantly higher intensities of these peaks were observed on the surface of spent Fe_3O_4@χ-Fe_5C_2 octahedra compared with spent nanocubes, suggesting a greater accumulation of deposited carbon on the octahedral surface (Supplementary Fig. 21a, b). For a more precise comparison, we conducted thermogravimetric analysis (TGA) under the N_2 atmosphere on both Fe_3O_4@χ-Fe_5C_2 nanocubes and octahedra after the reaction. The weight loss between 200 and 500 °C was attributed to the removal of long-chain hydrocarbons from the surface, while the weight loss beyond 500 °C was associated with the loss of non-graphitic carbon. In the case of spent Fe_3O_4@χ-Fe_5C_2 nanocubes, a weight loss of 6.2 wt% was observed (Supplementary Fig. 21c). Conversely, during TGA testing, spent Fe_3O_4@χ-Fe_5C_2 octahedra exhibited a total weight loss of 13.5 wt% due to long-chain hydrocarbons and deposited carbon (Supplementary Fig. 21d). The higher residual weight of long-chain hydrocarbons and deposited carbon on the octahedral structure suggests a more pronounced blockage of active sites compared to nanocubes. Hence, we hypothesize that carbon deposition also contributes to the deactivation of Fe_3O_4@χ-Fe_5C_2 octahedra.

Mechanistic insights into the facet effect

To rationalize the facet-dependent FTS activity, we explored CO dissociation pathways by conducting in-situ diffuse reflection infrared spectroscopy (DRIFTS) measurements at 270 °C. When Fe_3O_4@χ-Fe_5C_2 nanocubes were exposed to 1 bar of CO and purged with He, three sets of peaks were observed (Fig. 4c). The peaks at 2173 and 2115 cm^{-1} were assigned to the gaseous CO, while the peak at 2058 cm^{-1} arose from the stretching vibration of linearly adsorbed CO[19,40–42]. The peaks at 2360 and 2336 cm^{-1} corresponded to the gaseous CO_2, indicating the direct dissociation of CO on Fe_3O_4@χ-Fe_5C_2 nanocubes. As for the exposure to CO and purging with H_2, other sets of peaks emerged besides the peaks for CO and gaseous CO_2 (Fig. 4c). Specifically, the peak at 2955 cm^{-1} derived from the asymmetrical stretching vibration of C-H in CH_3* (refs. 43,44). The peaks at 2919 and 2849 cm^{-1} were indexed as the asymmetrical and symmetrical stretching vibrations of C-H in CHO* and CH_2*, respectively[45,46]. Notably, the peak at 1741 cm^{-1} was ascribed to the stretching vibration of C = O in CHO* species, which indicated the existence of a hydrogen-assisted dissociation route[47,48]. For Fe_3O_4@χ-Fe_5C_2 octahedra, when the sample was exposed to CO and purged with He, the peaks for gaseous CO_2 appeared (Fig. 4d). As for the treatment with CO and purging with H_2, the in-situ DRIFTS profile showed the peaks for CH_3*, CH_2*, CO_2, and CO, in the absence of CHO* or COH* (Fig. 4d). We also conducted in-situ DRIFTS experiments under 20 bar of syngas, simulating realistic reaction environments (Supplementary Fig. 22, Supplementary Table 5). The appearance of gaseous CO_2 peaks at 2360 and 2336 cm^{-1} provided evidence for direct dissociation occurring on both Fe_3O_4@χ-Fe_5C_2 nanocubes and octahedra (Supplementary Fig. 22, Supplementary Table 5). Moreover, the presence of CHO* species, as indicated by a peak at 1741 cm^{-1}, was observed solely on the Fe_3O_4@χ-Fe_5C_2 nanocubes catalyst and absent on the octahedral counterpart (Supplementary Fig. 22, Supplementary Table 5). Therefore, Fe_3O_4@χ-Fe_5C_2 octahedra enabled the direct dissociation of CO, while both direct and hydrogen-assisted CO dissociation routes existed on Fe_3O_4@χ-Fe_5C_2 nanocubes (Supplementary Fig. 23).

To investigate the activation energy of different CO dissociation routes, we plotted the area of linearly adsorbed CO peak at 2058 cm^{-1} as a function of time at 190, 230, and 270 °C (Supplementary Figs. 24–27). Based on the slope of the decrease in peak area with purge time, the rate of CO dissociation (k) was obtained. Activation energies of CO dissociation were calculated with the Arrhenius equation[7]. For Fe_3O_4@χ-Fe_5C_2 nanocubes, the activation energy for hydrogen-assisted dissociation of CO was 68.6 kJ mol^{-1}, significantly lower than that (103.7 kJ mol^{-1}) direct dissociation of CO (Supplementary Fig. 28a). With regard to Fe_3O_4@χ-Fe_5C_2 octahedra, the activation energy (83.2 kJ mol^{-1}) of CO dissociation with H_2 approximated to that (80.4 kJ mol^{-1}) without H_2 (Supplementary Fig. 28b). Therefore, Fe_3O_4@χ-Fe_5C_2 nanocrystals exhibited facet-dependent FTS activities derived from the alteration of reaction paths which changed the activation energy. Specially, Fe_3O_4@χ-Fe_5C_2 nanocubes followed hydrogen-assisted CO dissociation which lowered the activation energy, relative to that of direct CO dissociation over Fe_3O_4@χ-Fe_5C_2 octahedra.

We conducted DFT calculations to rationalize CO direct dissociation route and hydrogen-assisted dissociation path on χ-Fe_5C_2(202) and χ-Fe_5C_2(112) surface. For χ-Fe_5C_2(202) surface, the hydrogen-assisted CO dissociation route is the dominating route, since its energy barrier of the rate-limiting step (CO* + H* → HCO*) 1.23 eV is much lower than the direct CO dissociation route with an energy barrier as high as 2.85 eV (Supplementary Figs. 29–32). For χ-Fe_5C_2(112) surface, the direct CO dissociation route exhibited a lower energy barrier (1.37 eV) than the hydrogen-assisted CO dissociation route (1.54 eV), implying the direct dissociation as the main route of CO dissociation (Supplementary Figs. 33–36). Besides, the energy barrier of the hydrogen-assisted CO dissociation route on the χ-Fe_5C_2(202) surface is lower than that of the CO direct dissociation on the χ-Fe_5C_2(112) surface (Supplementary Figs. 32 and 35). The DFT conclusion was consistent with in-situ DRIFTS spectra result.

We explored the adsorption of CO and H_2 by conducting pulse chemisorption measurements to rationalize why Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC exhibited better carbon-chain growth ability. The amount of adsorbed gas was calculated on the difference between the total amount of gas injected and the amount measured at the outlet from the sample. The amount of adsorbed CO over Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was 24.1 μmol g^{-1}, higher than that (19.5 μmol g^{-1}) of Fe_3O_4@χ-Fe_5C_2 octahedra (Supplementary Fig. 12). While the amount of adsorbed H_2 over Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was 7.3 μmol g^{-1}, lower than that (11.0 μmol g^{-1}) of the octahedral counterpart (Supplementary Fig. 37). The results indicated that Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC improved the CO adsorption and suppressed the H_2 adsorption compared to the octahedral counterpart. Thus, the surface CO/H_2 ratio of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was 3.3, higher than that (1.8) of the octahedral counterpart (Supplementary Fig. 38). With the increase of the surface CO/H_2 ratio, the CH_4 production over Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC was suppressed, while hydrocarbon products shifted to long-chain hydrocarbons. We also conducted DFT

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calculations to investigate the facet effect on CH_4 production. The $CH_2^* + H^*$ energy barrier of χ-Fe_5C_2(202) is 1.03 eV, higher than that (0.70 eV) of χ-Fe_5C_2(112) (Supplementary Figs. 39–41). The energy barrier of the $CH_2^* + CH_2^*$ step over χ-Fe_5C_2(202) facet is 0.70 eV, lower than that (1.03 eV) of $CH_2^* + H^*$. As for χ-Fe_5C_2(112) facet, the energy barrier of the $CH_2^* + CH_2^*$ step is 0.60 eV, approaching to that (0.7 eV) of $CH_2^* + H^*$. Thus, compared with Fe_3O_4@χ-Fe_5C_2 nanocubes, the octahedral counterpart benefits the CH_4 production.

We demonstrated the dependence of catalytic performance on the exposed facet of iron carbides in FTS. Uniformly exposed {202} and {112} facets of χ-Fe_5C_2 were successfully fabricated as the model system. With the help of well-defined catalysts, we identified the intrinsically active, selective, and stable facets of χ-Fe_5C_2. Our findings deepen the understanding of Fe-based FTS catalysts from the phase and size to the facet. In addition, this work also provides a facile method to precisely control the exposed surfaces of iron carbides for both future fundamental studies and practical applications.

Methods

Chemicals and materials

Oleylamine (OAm, >70%), oleic acid (OA, 90%), benzyl ether (BE, 99%), iron(III) acetylacetonate (Fe(acac)$_3$, >99.9%), and 4-biphenylcarboxylic acid (99%) were obtained from Sigma-Aldrich. SiC (β-phase, 99.8%) was obtained from Adamas-beta®. All other chemicals were analytical grade and purchased from Sinopharm Chemical Reagent Co., Ltd. Deionized water with a resistivity of 18.2 MΩ cm was used for the preparation of all aqueous solutions (Milli-Q®).

Synthesis of Fe_3O_4 nanocrystals

In a typical synthesis of Fe_3O_4 nanocubes, Fe(acac)$_3$ (1.4 g) and 4-biphenylcarboxylic acid (1.0 g) were dissolved in a mixture of BE (20.0 mL) and OA (2.5 mL). The solution was degassed at 120 °C for 30 min. Subsequently, the solution was heated to 290 °C at 10 °C min^{-1} and kept for 30 min with a stirring rate of 300 rpm. As for the synthesis of Fe_3O_4 octahedra, Fe(acac)$_3$ (1.0 g) was dissolved in a mixture of BE (20.0 mL), OAm (2.3 mL), and OA (1.6 mL), followed by degassing at 120 °C for 30 min. The solution was heated to 220 °C at 20 °C min^{-1} with a stirring rate of 300 rpm and kept for 1 h. Afterward, the solution was heated to 300 °C at 20 °C min^{-1} and kept for 2 h. After the solution had been cooled down to room temperature, the products were precipitated by ethanol, washed three times with hexane, and re-dispersed in hexane.

Synthesis of Fe_3O_4@χ-Fe_5C_2 nanocrystals

For the synthesis of Fe_3O_4@χ-Fe_5C_2 nanocrystals, as-prepared Fe_3O_4 nanocrystals (50 mg) were loaded into a fixed-bed reactor with an inner diameter of 9 mm. Fe_3O_4 nanocrystals were reduced in H_2 under 1 bar with a gas-flow rate of 100 mL min^{-1} at 270 °C for 10 h, while the heating ramp was 1 °C min^{-1}. Afterwards, the obtained samples underwent surface reconstruction under 20 bar of syngas (32 vol% H_2, 64 vol% CO, and 4 vol% Ar) with a space velocity of 2400 mL h^{-1} g_{cat}^{-1} at 270 °C for 20 h. Considering that the iron carbides were highly sensitive to the atmosphere, we used inert gas to protect and store the catalyst after reaction. Specifically, we switched the feed gas to N_2 when the reaction was stopped. The reaction tubes were sealed via valves at both ends after cooling the reactor to room temperature. Afterwards, the reaction tubes were transferred to a N_2-filled glove box. The samples were stored in the glove box before characterizations.

Synthesis of pure χ-Fe_5C_2 nanocrystals

Octadecylamine of 14.5 g and cetyl trimethyl ammonium bromide (CTAB) of 0.113 g were mixed in a four-neck flask under stirring and degassed under the N_2 flow, followed by being heated to 120 °C. Afterwards, Fe(CO)$_5$ (3.6 mmol) was injected into the mixture under the N_2 blanket. The mixture was heated to 180 °C at 10 °C min^{-1} and kept at this temperature for 10 min. Subsequently, the mixture was further heated to 350 °C at 10 °C min^{-1} and kept at this temperature for 10 min. After being cooled to room temperature, the products were washed with a mixture of ethanol and hexane.

Mössbauer measurements

^{57}Fe Mössbauer spectra were carried out on a Topologic 500 A spectrometer driving with a proportional counter at room temperature. The radioactive source was ^{57}Co (Rh) moving in a constant acceleration mode. Data analyses were performed assuming a Lorentzian lineshape for computer folding and fitting.

Catalytic tests

The Fischer-Tropsch reaction was carried out in a fixed-bed reactor under 20 bar of syngas at 270 °C. Generally, Fe_3O_4@χ-Fe_5C_2 nanocrystals (50 mg, 20–40 meshes) were diluted with SiC (450 mg, 20–40 meshes). The sample was loaded into a fixed-bed reactor with an inner diameter of 9 mm. Subsequently, a mixture including 96 vol% of H_2/CO mixed gas (64 vol% H_2, 32 vol% CO) and 4 vol% of Ar (as an internal standard) was introduced to the reactor as the feeding gas at a space velocity of 2400 mL h^{-1} g_{cat}^{-1}.

The gaseous products were monitored by online gas chromatographs (Shimadzu GC-2014). An ice trap with 2.0 g of solvent (*n*-dodecane) was employed to capture the liquid hydrocarbons in the effluent. The liquid products with 0.1 g of an internal standard (decalin) were analyzed using an offline Shimadzu GC-2014.

CO conversion was calculated as follows:

$$\text{CO conversion} = \frac{CO_{inlet} - CO_{outlet}}{CO_{inlet}} \times 100\% \quad (3)$$

where CO_{inlet} and CO_{outlet} are moles of CO at the inlet and outlet, respectively.

CO_2 selectivity was calculated according to:

$$CO_2\ \text{selectivity} = \frac{CO_{outlet}}{CO_{inlet} - CO_{outlet}} \times 100\% \quad (4)$$

where $CO_{2\ outlet}$ refers to moles of CO_2 at the outlet.

The selectivity of hydrocarbon C_nH_m was obtained according to:

$$C_nH_m\ \text{selectivity} = \frac{nC_nH_{m\,outlet}}{\sum_i iC_iH_{m\,outlet}} \times 100\% \quad (5)$$

where $C_nH_{m\,outlet}$ represents moles of individual hydrocarbon product at the outlet.

Carbon balance was calculated according to:

$$\text{Carbon balance} = \frac{\sum_i iC_iH_{m\,outlet} + CO_2}{CO_{inlet} - CO_{outlet}} \times 100\% \quad (6)$$

where C_nH_m and CO_2 represent the moles of the produced hydrocarbons and CO_2, respectively. CO_{inlet} and CO_{outlet} are moles of CO at the inlet and outlet, respectively. The carbon balances were over 95.0%.

In-situ DRIFTS measurements after different gas treatments

In-situ DRIFTS experiments were conducted in an elevated-pressure cell (DiffusIR Accessory PN 041-10XX) with a Fourier transform infrared spectrometer (TENSOR II Sample Compartment RT-DLaTGS) and a liquid-nitrogen-cooled MCT detector. The outlet gas was analyzed by a mass spectrometer (Hiden HPR20). Spectra were measured accumulating 64 scans at a resolution of 4 cm^{-1}. Prior to the test, the samples were cleaned in He with a gas-flow rate of 100 mL min^{-1} at 270 °C for 1 h. Then the samples were exposed to 1 bar of CO for 30 min and subsequently purged by He or H_2 for 30 min at 270 °C. As for the in-situ DRIFTS measurements at different temperatures, the background spectra of the samples were acquired under He after the temperature of the samples dropped to a specified temperature. Then, the samples were exposed to 1 bar of CO for 30 min and subsequently purged with He or H_2 at a specified temperature. As for the in-situ DRIFTS experiments under 20 bar of syngas, thorough cleaning of the samples in He at 270 °C for 1 h ensured the elimination of any contaminants. Background spectra were acquired under He flow, followed by exposure to a 20-bar syngas mixture (CO/H_2) at 270 °C for 30 min.

DFT methods

Spin-polarized DFT calculations were conducted using the Vienna ab initio simulation package (VASP)[49]. The projector-augmented wave method and the Perdew-Burke-Ernzerhof functional were implemented in the code[50]. Fe_5C_2(202) and Fe_5C_2(112) slab were established with the same stoichiometric atoms ($Fe_{60}C_{24}$). A plane-wave basis set with a cutoff energy of 400 eV was used with the K-points of 3 × 3 × 1 and vacuum thickness of 15 Å. During the structure optimization, the two layers of atoms at the bottom were fixed, while the others include the adsorbates were relaxed with the electronic convergence of 0.02 eV/Å. For each elementary step, the initial states and final states are firstly optimized. The transition states are searched using the climbing image nudged elastic band method (CI-NEB) and confirmed by the vibrational frequencies analysis[51].

CO pulse chemisorption measurements

CO pulse chemisorption measurements were performed using a Micromeritics Autochem 2920 chemisorption analyzer with an active loop volume of 0.1 mL. In a typical measurement, 100 mg of Fe_3O_4@χ-Fe_5C_2 nanocrystals/SiC were packed into a reactor with a quartz tube. Prior to the test, the samples were cleaned in He with a gas-flow rate of 100 mL min^{-1} at 270 °C for 5 h. After cooling down to 50 °C under He flow, CO/He pulses (10 vol% CO and 90 vol% He) were injected until adsorption reached saturation. The amount of adsorbed CO was calculated on the difference between the total amount of CO injected and the amount measured at the outlet from the sample. The metal dispersion was calculated by assuming the ratio of CO to surface metal atom as 1:1.

TOF number was calculated according to:

$$\begin{aligned} \mathrm{TOF} = {} & \mathrm{CO\ conversion} \times \mathrm{moles\ of\ CO\ in\ syngas} \times \mathrm{gas} \\ & - \mathrm{flow\ rate} \div \mathrm{moles\ of\ surface\ Fe\ atoms} \end{aligned} \tag{7}$$

H_2 pulse chemisorption measurements

H_2 pulse chemisorption measurements were performed using a Micromeritics Autochem 2920 chemisorption analyzer with an active loop volume of 0.1 mL. In a typical measurement, 100 mg of Fe_3O_4@χ-Fe_5C_2 nanocrystals/SiC were packed into a reactor with a quartz tube. Prior to the test, the samples were cleaned in He with a gas-flow rate of 100 mL min^{-1} at 270 °C for 5 h. After cooling down to 50 °C under He flow, H_2/Ar pulses (10 vol% H_2 and 90 vol% Ar) were injected until adsorption reached saturation. The amount of adsorbed H_2 was calculated on the difference between the total amount of H_2 injected and the amount measured at the outlet from the sample.

Instrumentations

TEM images were taken using a Hitachi H-7700 transmission electron microscope at an acceleration voltage of 100 kV. HAADF and EDS analysis were collected on a JEOL ARM-200F field-emission transmission electron microscope operating at 200 kV accelerating voltage. XPS measurements were conducted on an ESCALAB 250 (Thermo-VG Scientific, USA) with an Al Kα X-ray source (1486.6 eV protons) in Constant Analyzer Energy (CAE) mode with a pass energy of 30 eV for all spectra. XRD characterization was performed using a Philips X'Pert Pro X-ray diffractometer with a monochromatized Cu Kα radiation source and a wavelength of 0.1542 nm. BET measurements were carried out on Micromeritics AutoChem II 2020. TGA spectra were conducted on Pyris Diamond TG-DTG. Raman spectra were detected by a Renishaw RM3000 Micro-Raman system with a 514.5 nm Ar laser.

Data availability

The data generated in this study are provided in the Supplementary Information.

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Author contributions

W.W. and J.L. equally contributed to this work. W.W., J.L., H.L., and J. Zeng designed the studies and wrote the paper. W.W., J.L., and L.L. synthesized catalysts. W.W., J.L., B.H., and H.L. performed catalytic tests. W.W. and W.M. conducted characterizations. J. Zhao conducted DFT calculations. S.H. and C.M. conducted HAADF-STEM characterizations. All authors discussed the results and commented on the paper.

Competing interests

The authors declare no competing interests.

Additional information

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Supplementary information for

Facet sensitivity of iron carbides in Fischer-Tropsch synthesis

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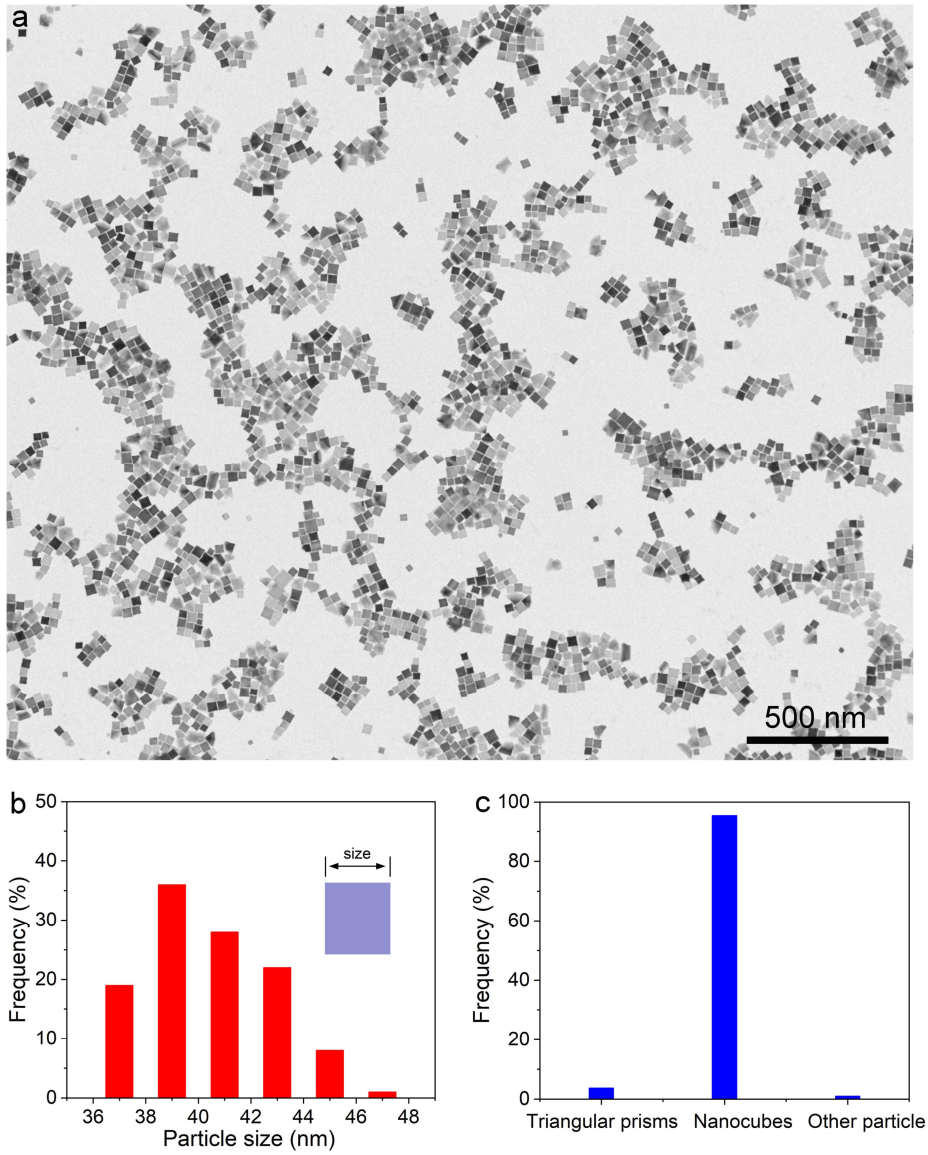
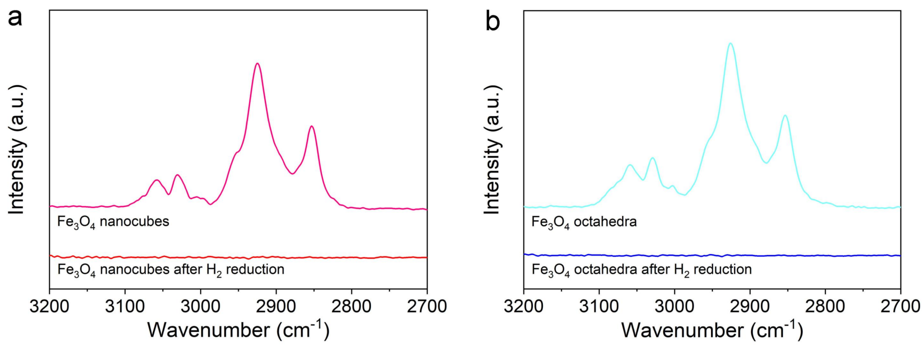
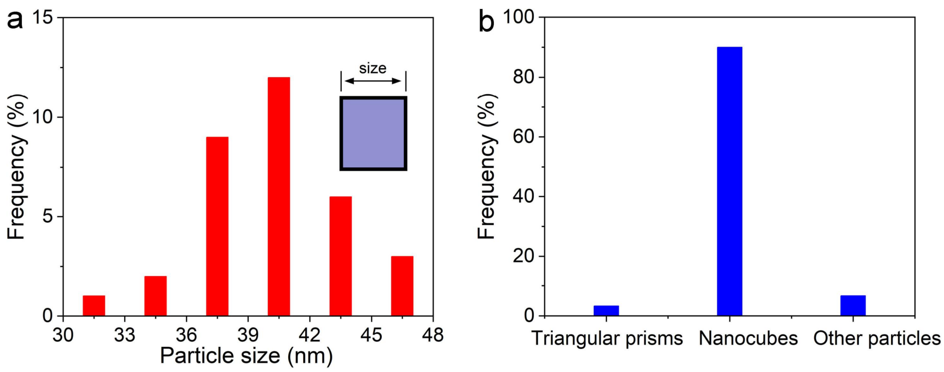


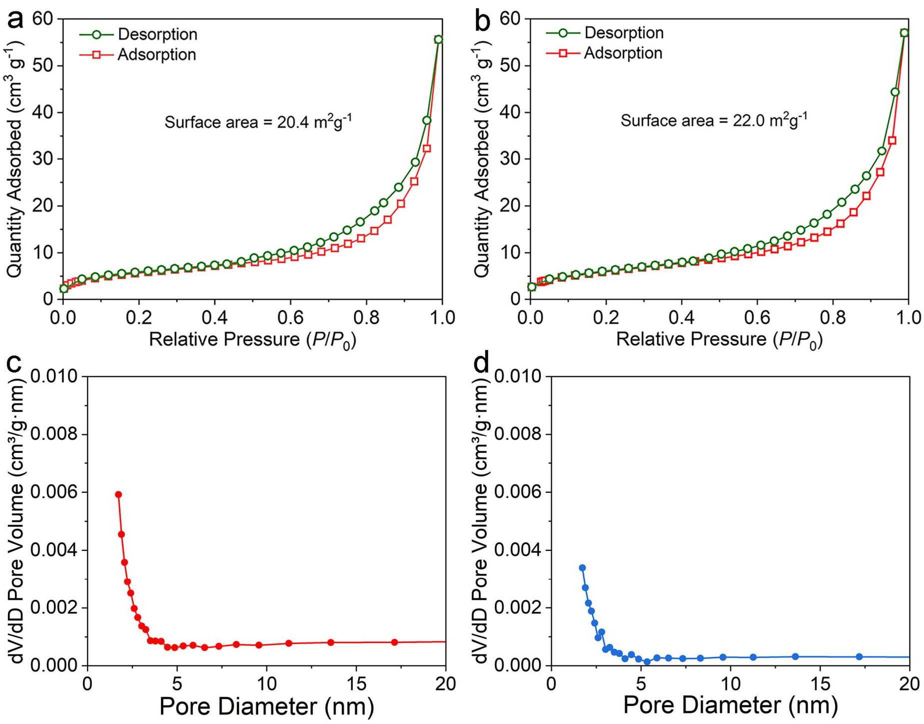
Supplementary Figure 1 | Structural characterizations of Fe_3O_4 nanocubes. (**a**) TEM image of Fe_3O_4 nanocubes. (**b**) Size distribution of the corresponding Fe_3O_4 nanocubes. The average particle size was 40.5±3.9 nm. (**c**) Contents of nanocrystals with different shapes. The purity of Fe_3O_4 nanocubes was 95.3%.



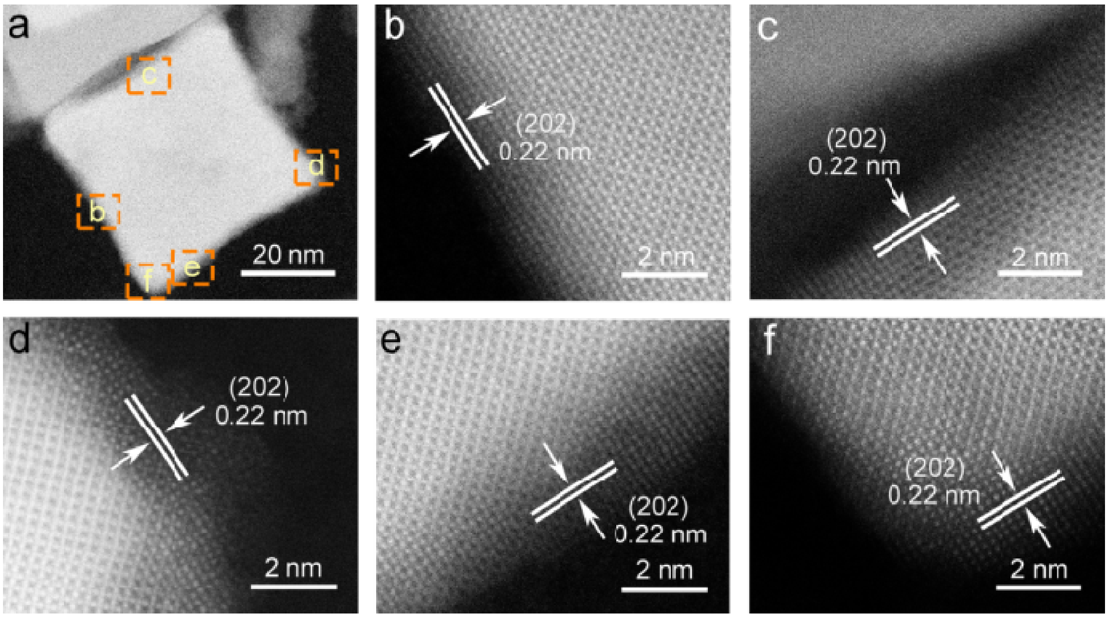
Supplementary Figure 2 | FTIR spectra. (**a**) FTIR spectra of Fe_3O_4 nanocubes before/after H_2 reduction. (**b**) FTIR spectra of Fe_3O_4 octahedra before/after H_2 reduction.



Supplementary Figure 3 | Statistics of Fe_3O_4@χ-Fe_5C_2 nanocubes from Figure 1a. (**a**) Size distribution of Fe_3O_4@χ-Fe_5C_2 nanocubes. The average particle size was 40.4±3.8 nm. (**b**) Contents of nanocrystals with different shapes. The purity of Fe_3O_4@χ-Fe_5C_2 nanocubes was 90.0%.



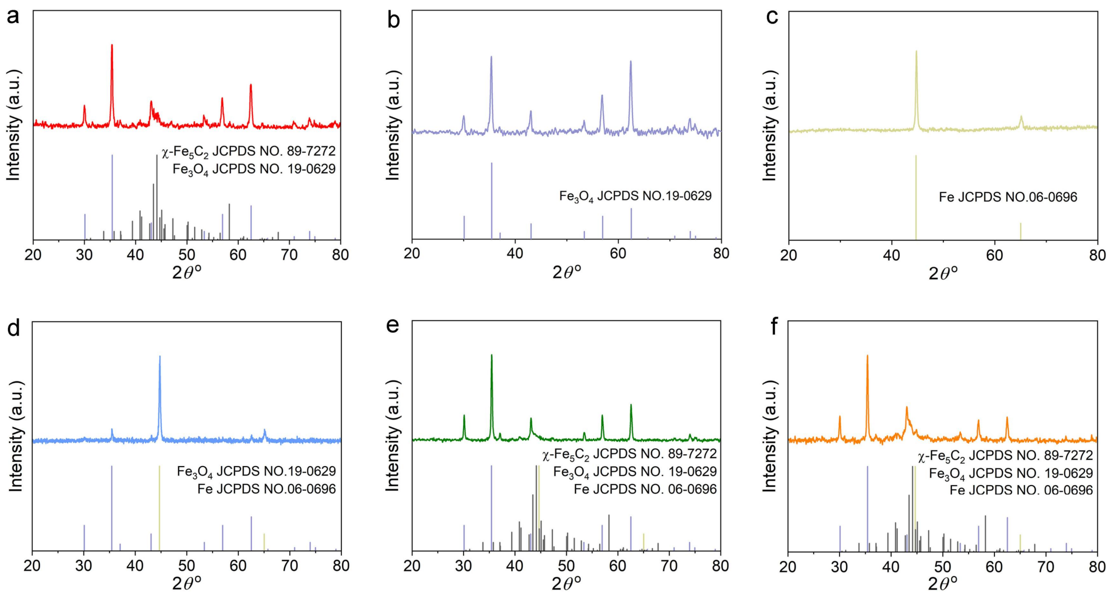
Supplementary Figure 4 | Textural properties of Fe_3O_4@χ-Fe_5C_2 nanocubes and octahedra before reaction. (**a, b**) Nitrogen adsorption and desorption isotherm of (**a**) Fe_3O_4@χ-Fe_5C_2 nanocubes and (**b**) Fe_3O_4@χ-Fe_5C_2 octahedra. (**c, d**) Pore-size distributions of (**c**) Fe_3O_4@χ-Fe_5C_2 nanocubes and (**d**) Fe_3O_4@χ-Fe_5C_2 octahedra derived from the nitrogen adsorption-desorption isotherms by the BJH method.



Supplementary Figure 5 | Structural characterizations of the Fe_3O_4@χ-Fe_5C_2 nanocube. (a) HAADF-STEM image of an individual Fe_3O_4@χ-Fe_5C_2 nanocube. **(b-f)** Magnified HAADF-STEM images of the region marked by the corresponding boxes in panel **a**.

Supplementary Table 1 | Mössbauer parameters of Fe_3O_4@χ-Fe_5C_2 nanocubes and Fe_3O_4@χ-Fe_5C_2 octahedra. The isomer shift (IS), quadrupole splitting (QS), hyperfine field, and spectral contribution are given.

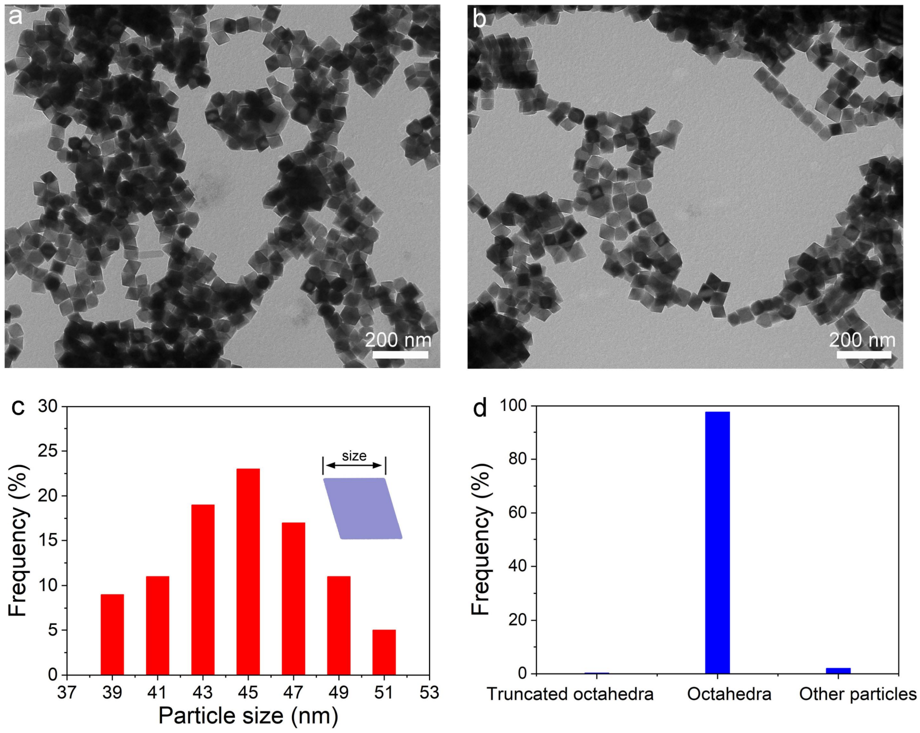
Samples	Phase ascription	Mössbaure parameters			
		IS (mm/s)	QS (mm/s)	Hyperfine field (T)	Spectral contribution (%)
Fe_3O_4@χ-Fe_5C_2 nanocubes	Fe_3O_4 (A)	0.28	-0.01	48.8	47.5%
	Fe_3O_4 (B)	0.65	0.01	45.7	15.9%
	χ-Fe_5C_2 (A)	0.22	0.10	21.9	11.0%
	χ-Fe_5C_2 (B)	0.20	-0.01	18.7	14.5%
	χ-Fe_5C_2 (C)	0.19	-0.07	11.1	7.7%
	Fe(II)/Fe(III)	0.28	1.06	-	3.4%
Fe_3O_4@χ-Fe_5C_2 octahedra	Fe_3O_4 (A)	0.32	-0.02	49.6	49.1%
	Fe_3O_4 (B)	0.65	0.02	45.8	16.0%
	χ-Fe_5C_2 (A)	0.24	0.10	21.7	13.5%
	χ-Fe_5C_2 (B)	0.19	-0.02	18.1	10.8%
	χ-Fe_5C_2 (C)	0.20	0.06	11.7	5.2%
	Fe(II)/Fe(III)	0.41	1.01	-	5.4%



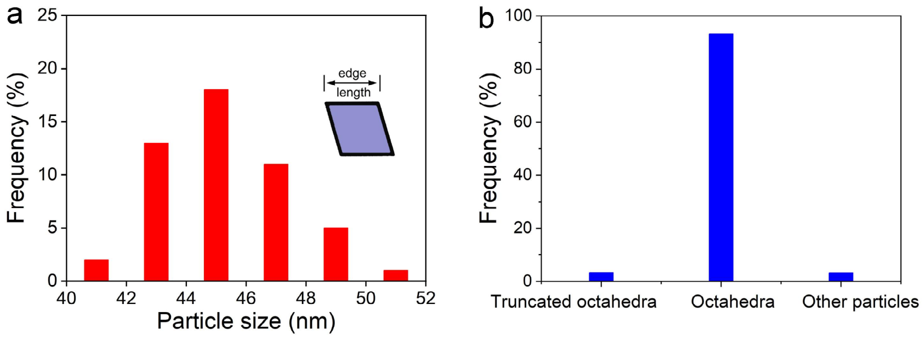
Supplementary Figure 6 | The corresponding XRD patterns of evolution from Fe_3O_4 nanocubes to Fe_3O_4@χ-Fe_5C_2 nanocubes in Figure 2. (**a**) XRD pattern of Fe_3O_4@χ-Fe_5C_2 nanocubes after the treatment of Fe nanocubes with syngas for 20 h. (**b**) XRD pattern of Fe_3O_4 nanocubes. (**c**) XRD pattern of Fe nanocubes. (**d-f**) XRD patterns of Fe_3O_4@χ-Fe_5C_2 nanocubes after the treatment of Fe nanocubes with syngas for 2, 5, and 10 h, respectively.

Supplementary Table 2 | Quantitative analysis of the XRD patterns. The calculation was based on the Ratio of Intensity Reference (RIR) method.

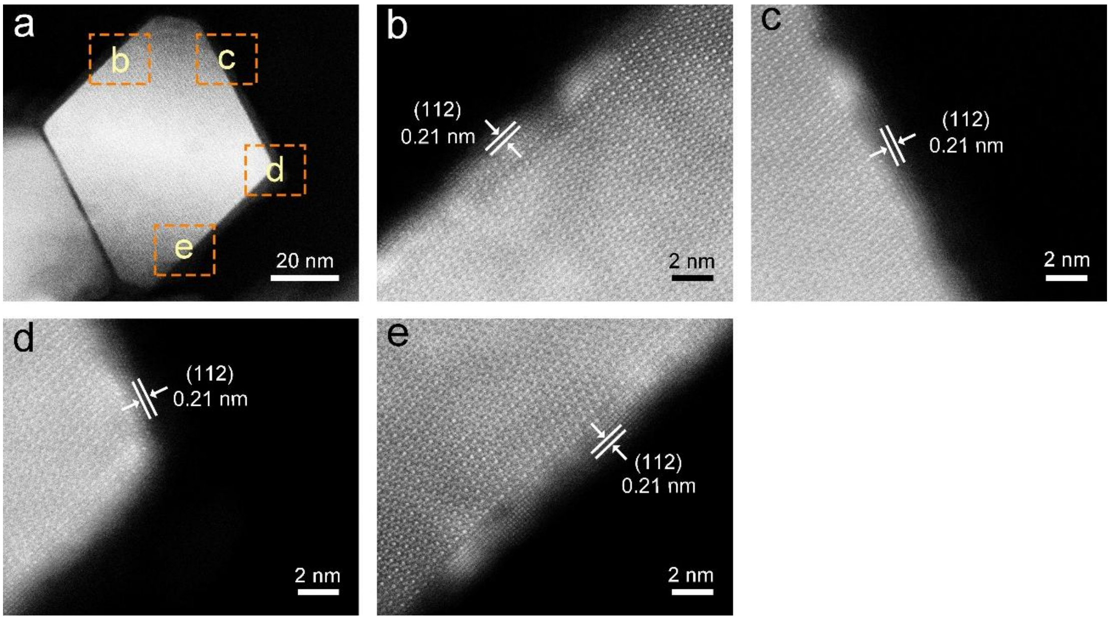
sample	Iron phase	Mass fraction (wt.%)
Fe_3O_4@χ-Fe_5C_2 nanocubes	Fe_3O_4	70.2
	χ-Fe_5C_2	29.8
Fe_3O_4@χ-Fe_5C_2 octahedra	Fe_3O_4	72.4
	χ-Fe_5C_2	27.6



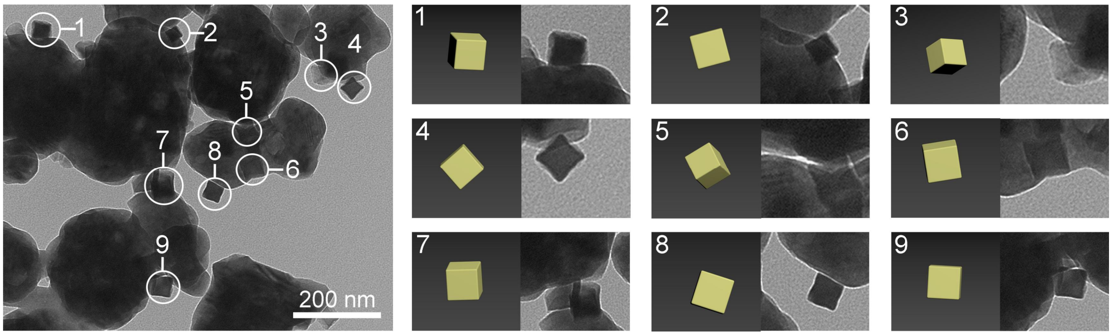
Supplementary Figure 7 | Structural characterizations of Fe_3O_4 octahedra. (**a-b**) TEM images of Fe_3O_4 octahedra. (**c**) Size distribution of the corresponding Fe_3O_4 octahedra. The average particle size was 44.8±3.3 nm. (**d**) Contents of nanocrystals with different shapes. The purity of Fe_3O_4 octahedra is 97.7%.



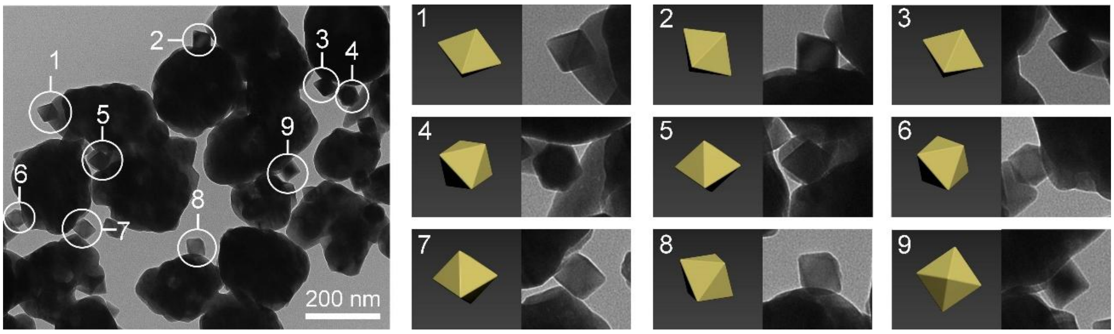
Supplementary Figure 8 | Statistics of Fe_3O_4@χ-Fe_5C_2 octahedra from Figure 3a. (**a**) Size distribution of Fe_3O_4@χ-Fe_5C_2 octahedra. The average particle size is 45.4±3.5 nm. (**b**) Contents of nanocrystals with different shapes. The purity of Fe_3O_4@χ-Fe_5C_2 octahedra is 93.3%.



Supplementary Figure 9 | Structural characterizations of an individual Fe_3O_4@χ-Fe_5C_2 octahedron/SiC. (**a**) HAADF-STEM image of an individual Fe_3O_4@χ-Fe_5C_2 octahedron/SiC. (**b-e**) Magnified HAADF-STEM images of the region marked by the corresponding boxes in panel **a**.



Supplementary Figure 10 | Structural characterizations of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC. TEM image of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC and the cubic models from different orientations.



Supplementary Figure 11 | Structural characterizations of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC. TEM image of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC and the octahedral models from different orientations.

Supplementary Table 3 | Catalytic properties of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC and octahedra/SiC towards FTS.

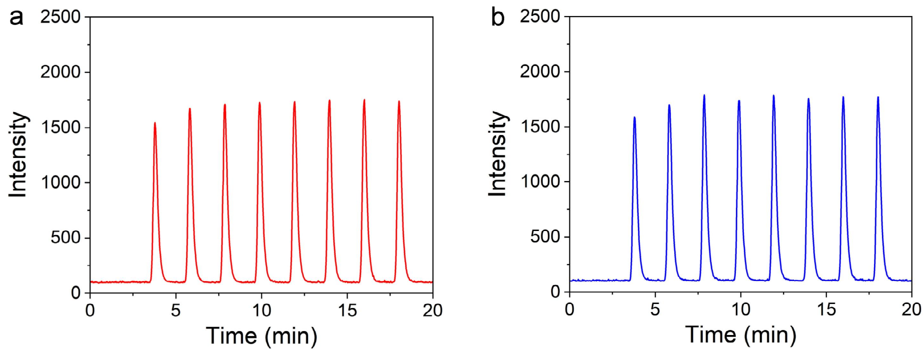
Catalysts	Conversion (%)	CO_2 selectivity (%)	Product selectivity (C%, CO_2-free)						Carbon balance (%)
			CH_4	C_2-$C_4^=$	C_2-C_4^o	C_5-$C_{12}^=$	C_5-C_{12}^o	C_{13+}	
Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC[a]	45.4	18.7	14.2	21.6	19.4	17.9	24.3	2.6	98.7
Fe_3O_4@χ-Fe_5C_2 octahedra/SiC[a]	21.2	12.8	19.3	20.6	32.5	11.3	14.5	1.8	96.5
χ-Fe_5C_2 nanoparticles/SiC[a]	23.6	10.5	15.5	29.8	18.4	16.9	18.0	1.4	96.8
Fe_3O_4@χ-Fe_5C_2 octahedra/SiC[b]	42.6	16.3	20.1	18.3	22.0	16.0	21.5	2.1	97.6

[a] refers to the conditions of 20 bar, syngas (64 vol% H_2, 32 vol% CO, and 4 vol% Ar), 2400 mL h^{-1} g_{cat}^{-1}, and 270 °C.

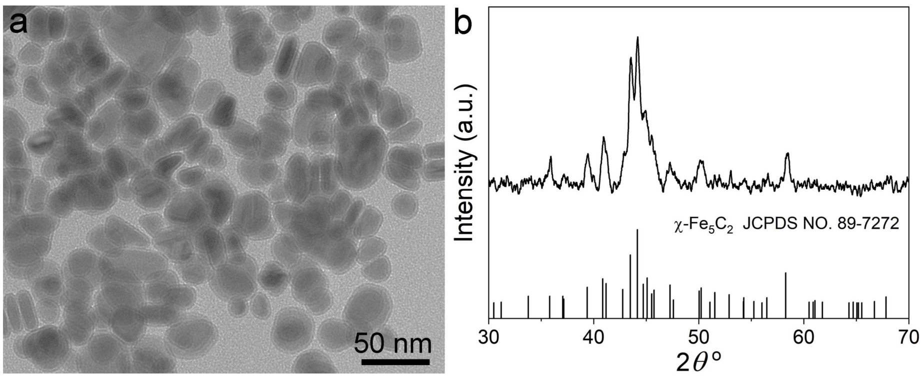
[b] refers to the conditions of 20 bar, syngas (64 vol% H_2, 32 vol% CO, and 4 vol% Ar), 800 mL h^{-1} g_{cat}^{-1}, and 270 °C.

$^=$ refers to olefins.

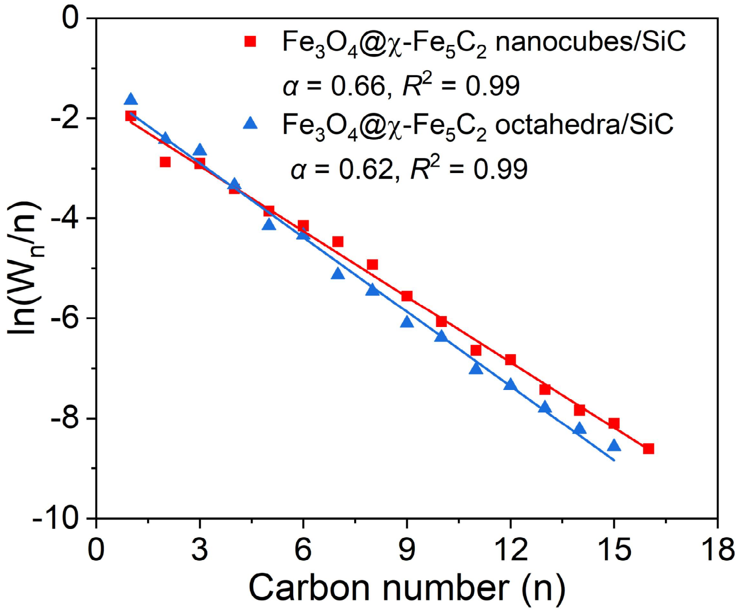
^{o} refers to paraffins.



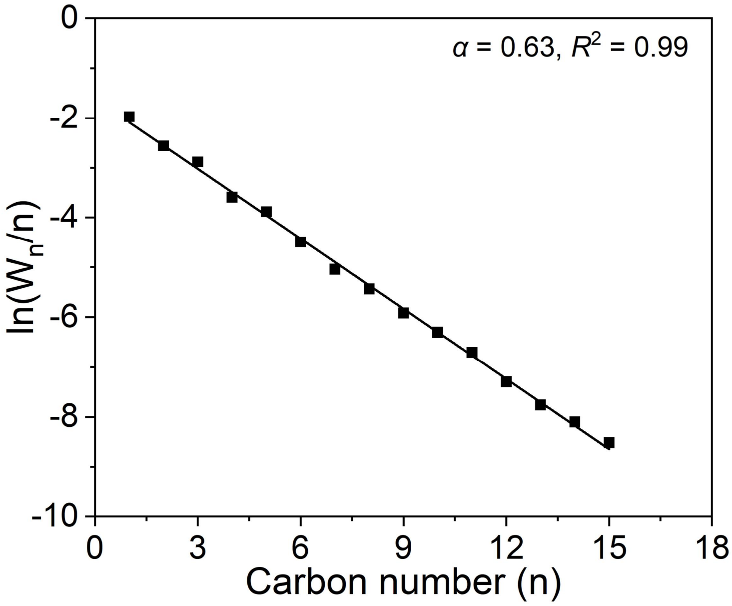
Supplementary Figure 12 | CO pulse profiles. CO pulse profiles of (**a**) Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC and (**b**) Fe_3O_4@χ-Fe_5C_2 octahedra/SiC.



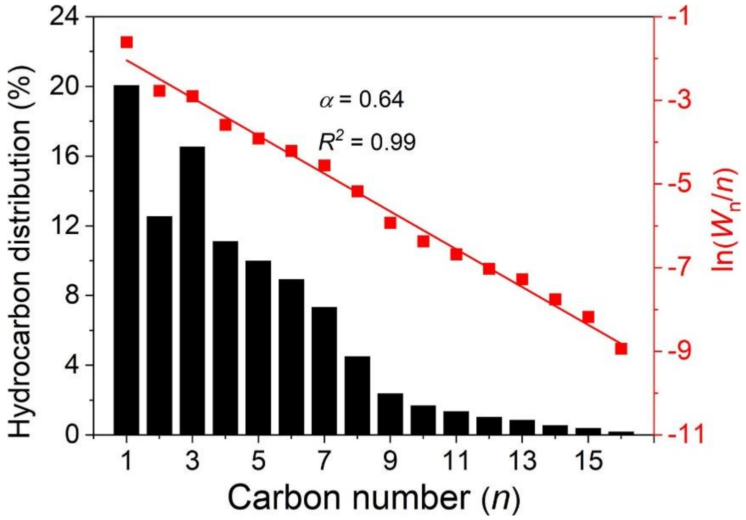
Supplementary Figure 13 | Structural characterizations of pure χ-Fe_5C_2 nanoparticles. (a) TEM image of pure χ-Fe_5C_2 nanoparticles. (**b**) XRD pattern of pure χ-Fe_5C_2 nanoparticles.



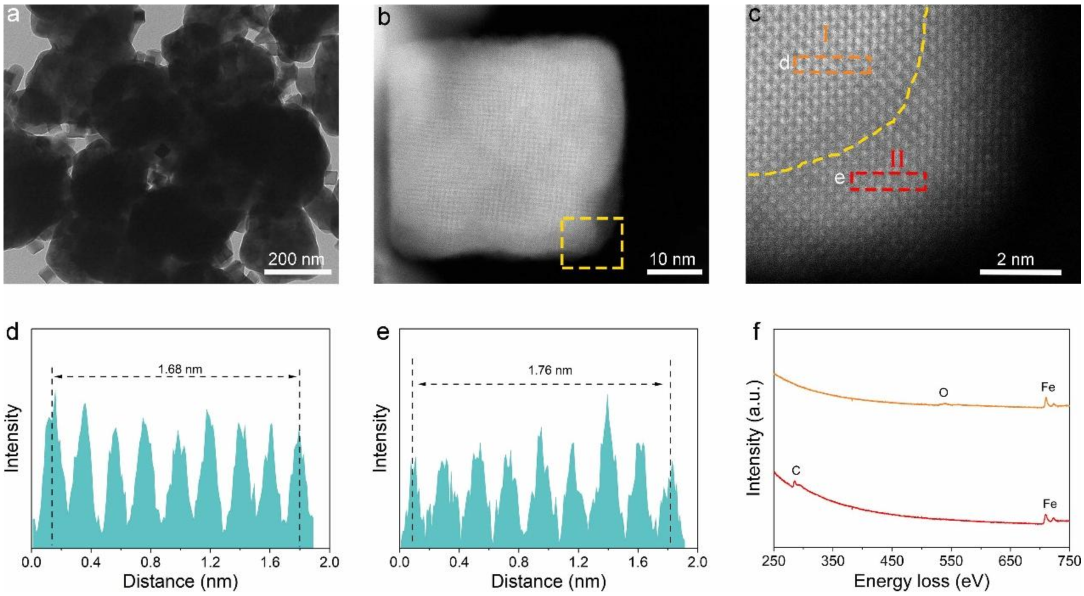
Supplementary Figure 14 | The ASF plot and the corresponding α value of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC and Fe_3O_4@χ-Fe_5C_2 octahedra/SiC. The reaction was conducted under 20 bar of syngas (CO:H_2 = 1:2, 2400 mL h^{-1} g_{cat}^{-1}) at 270 °C.



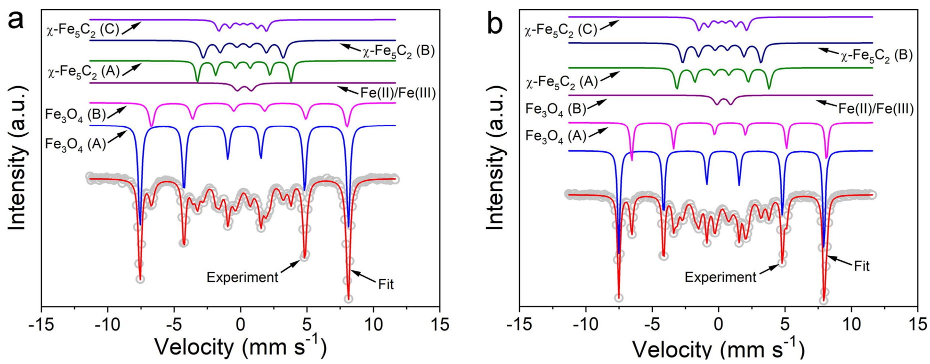
Supplementary Figure 15 | The ASF plot and the corresponding α value of χ-Fe_5C_2 nanoparticles/SiC. The reaction was conducted under 20 bar of syngas ($CO:H_2$ = 1:2, 2400 mL h^{-1} g_{cat}^{-1}) at 270 ^oC.



Supplementary Figure 16 | Hydrocarbon distribution, the ASF plot, and the corresponding *α* value of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC at similar conversion levels of Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC. The reaction was conducted under 20 bar of syngas (CO:H_2 = 1:2, 800 mL h^{-1} g_{cat}^{-1}) at 270 °C.



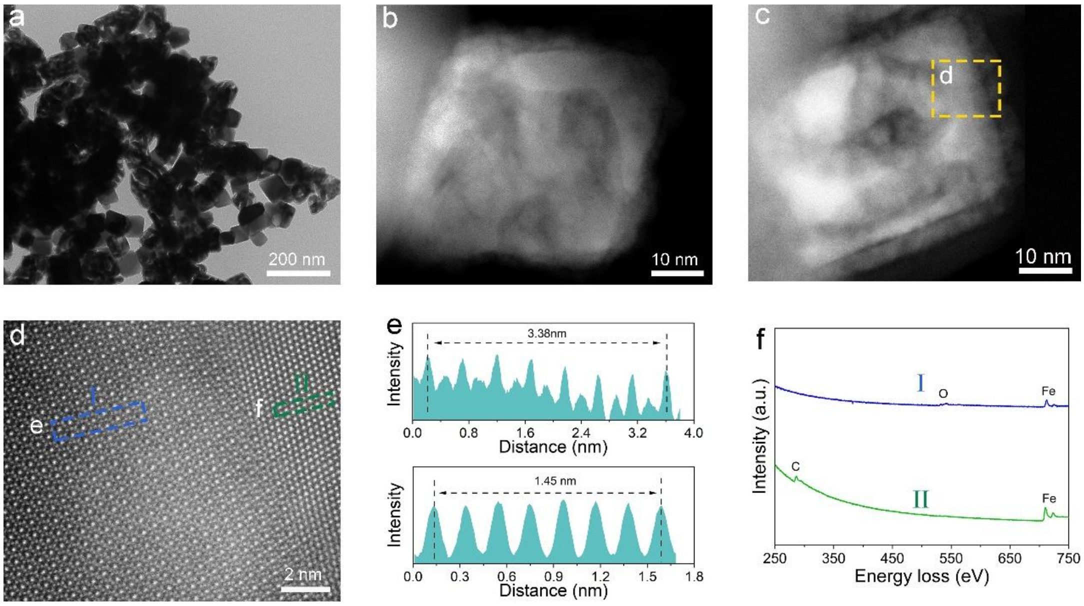
Supplementary Figure 17 | Structural characterizations of Fe_3O_4@χ-Fe_5C_2 nanocubes after 100 h on stream. (**a**) TEM image of Fe_3O_4@χ-Fe_5C_2 nanocube/SiC. (**b**) HAADF-STEM image of an individual Fe_3O_4@χ-Fe_5C_2 nanocube. (**c**) Magnified HAADF-STEM image of the region marked by the corresponding boxes in panel **b**. (**d**) Intensity profile recorded from the area indicated by the rectangular box in panel **c**. (**e**) Intensity profile recorded from the area indicated by the rectangular box in panel **c**. (**f**) EELS spectra of a Fe_3O_4@χ-Fe_5C_2 nanocube in panel **c**.



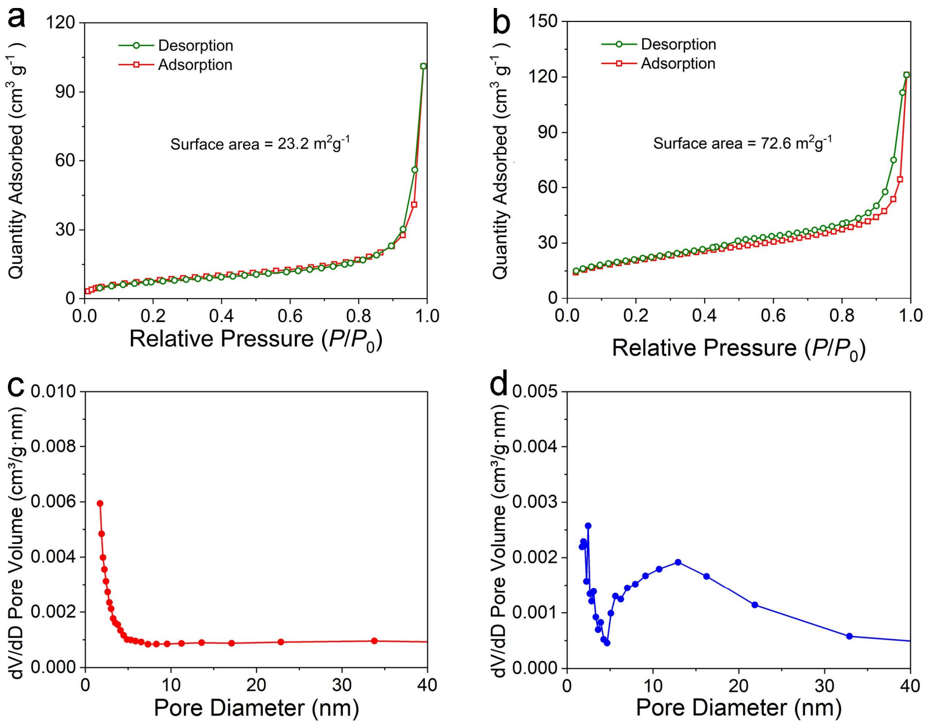
Supplementary Figure 18 | Mössbauer spectra characterizations. (**a**) Mössbauer spectra of Fe_3O_4@χ-Fe_5C_2 nanocubes after 100 h on stream. (**b**) Mössbauer spectra of Fe_3O_4@χ-Fe_5C_2 octahedra after 100 h on stream.

Supplementary Table 4 | Mössbauer parameters of Fe_3O_4@χ-Fe_5C_2 nanocubes and Fe_3O_4@χ-Fe_5C_2 octahedra after 100 h on stream. The isomer shift (IS), quadrupole splitting (QS), hyperfine field, and spectral contribution are given.

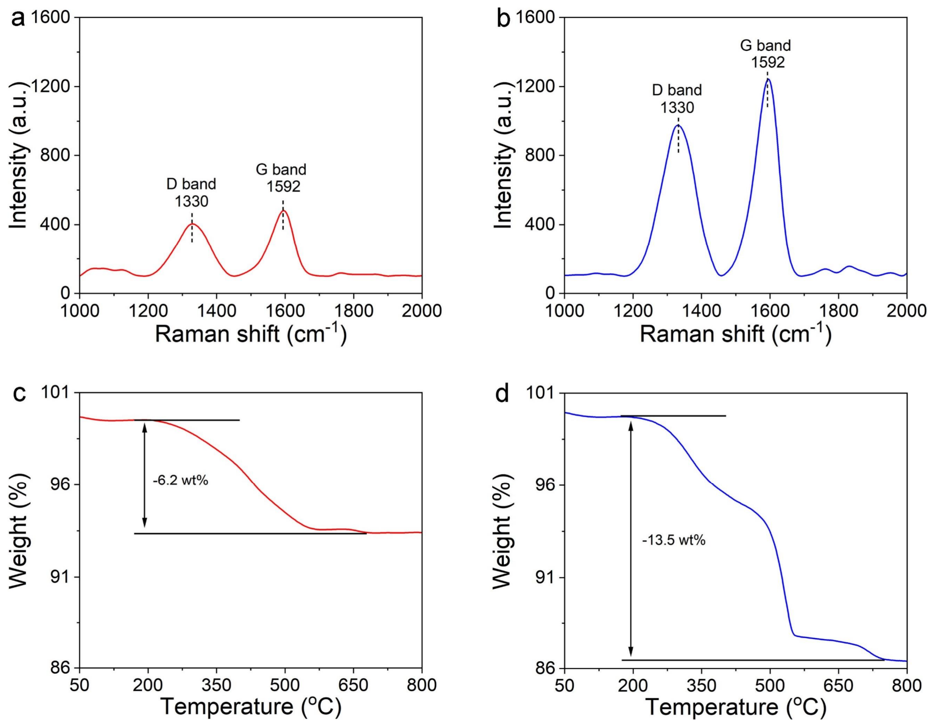
Samples	Phase ascription	Mössbaure parameters			
		IS (mm/s)	QS (mm/s)	Hyperfine field (T)	Spectral contribution
Fe_3O_4@χ-Fe_5C_2 nanocubes after reaction	Fe_3O_4 (A)	0.30	-0.01	48.9	40.7%
	Fe_3O_4 (B)	0.65	0.01	45.5	15.4%
	χ-Fe_5C_2 (A)	0.22	0.11	21.8	12.9%
	χ-Fe_5C_2 (B)	0.17	0.06	18.6	17.0%
	χ-Fe_5C_2 (C)	0.20	0.02	11.1	9.9%
	Fe(II)/Fe(III)	0.34	1.07	-	4.1%
Fe_3O_4@χ-Fe_5C_2 octahedra after reaction	Fe_3O_4 (A)	0.33	-0.05	48.5	38.7%
	Fe_3O_4 (B)	0.78	-0.09	45.4	16.6%
	χ-Fe_5C_2 (A)	0.27	0.11	21.6	14.5%
	χ-Fe_5C_2 (B)	0.22	0.04	18.3	15.5%
	χ-Fe_5C_2 (C)	0.28	0.06	11.2	10.4%
	Fe(II)/Fe(III)	0.39	1.04	-	4.3%



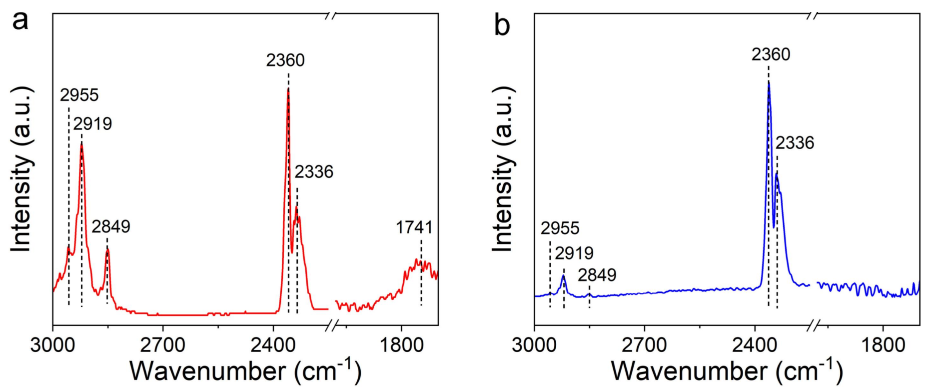
Supplementary Figure 19 | Structural characterizations of Fe_3O_4@χ-Fe_5C_2 octahedra after 100 h on stream. (**a**) TEM image of Fe_3O_4@χ-Fe_5C_2 octahedra/SiC. (**b**) HAADF-STEM image of an individual Fe_3O_4@χ-Fe_5C_2 octahedron. (**c**) HAADF-STEM image of another Fe_3O_4@χ-Fe_5C_2 octahedron. (**c**) Magnified HAADF-STEM image of the region marked by the corresponding boxes in panel **b**. (**d**) Intensity profile recorded from the area indicated by the rectangular box in panel **d**. (**e**) Intensity profile recorded from the area indicated by the rectangular box in panel **d**. (**f**) EELS spectra of a Fe_3O_4@χ-Fe_5C_2 octahedron in panel **d**.



Supplementary Figure 20 | Textural properties of Fe_3O_4@χ-Fe_5C_2 nanocubes and octahedra after 100 h on stream. (a, b) Nitrogen adsorption and desorption isotherm of (**a**) Fe_3O_4@χ-Fe_5C_2 nanocubes and (**b**) Fe_3O_4@χ-Fe_5C_2 octahedra after 100 h on stream. (**c, d**) Pore-size distributions of (**c**) Fe_3O_4@χ-Fe_5C_2 nanocubes and (**d**) Fe_3O_4@χ-Fe_5C_2 octahedra after 100 h on stream derived from the nitrogen adsorption-desorption isotherms by the BJH method.



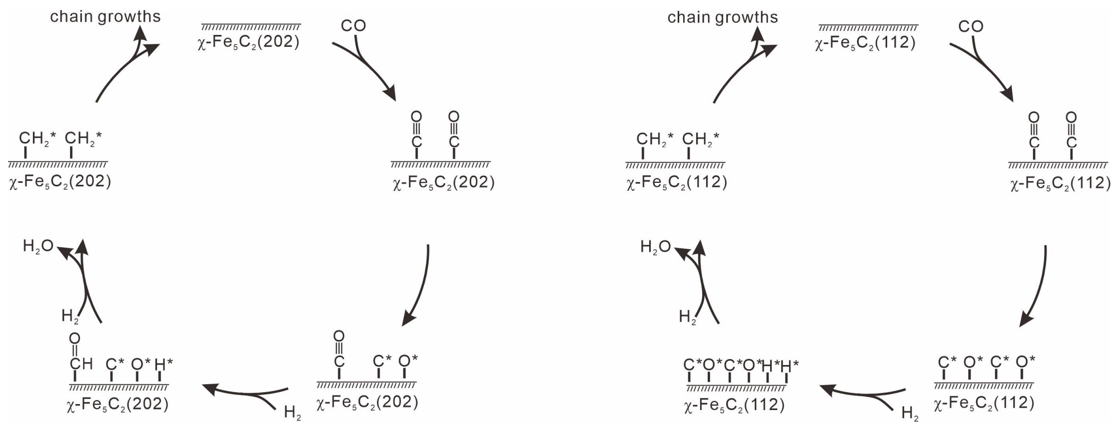
Supplementary Figure 21 | Raman spectra of Fe_3O_4@χ-Fe_5C_2 nanocubes (**a**) and Fe_3O_4@χ-Fe_5C_2 octahedra (**b**) after 100 h on stream. TGA profiles of Fe_3O_4@χ-Fe_5C_2 nanocubes (**c**) and Fe_3O_4@χ-Fe_5C_2 octahedra (**d**) after 100 h on stream.



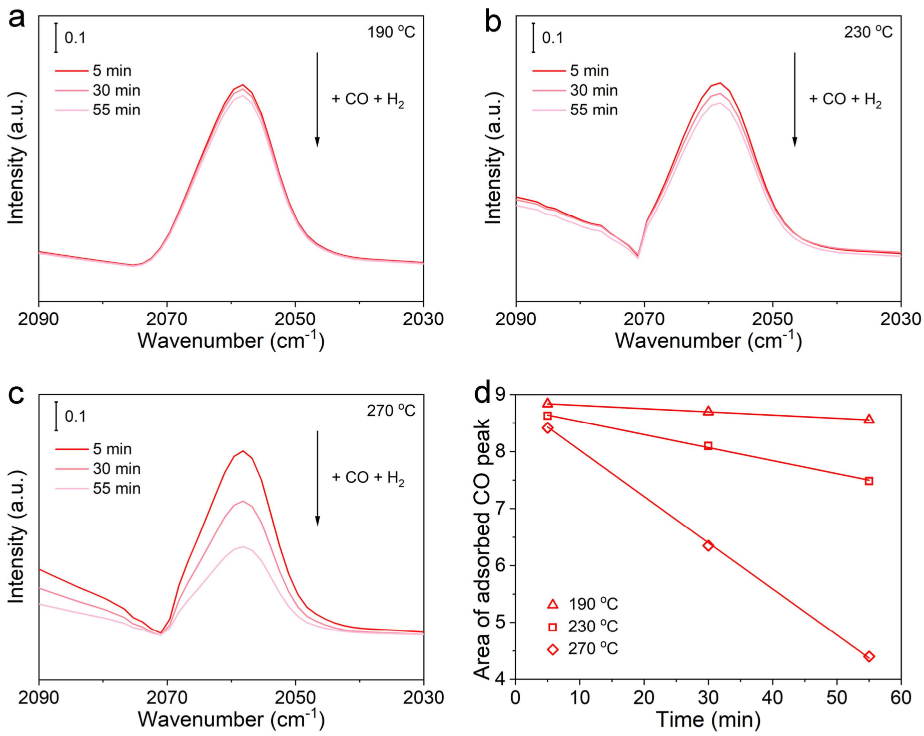
Supplementary Figure 22 | (a) *In-situ* DRIFTS spectra of Fe_3O_4@χ-Fe_5C_2 nanocubes and **(b)** Fe_3O_4@χ-Fe_5C_2 octahedra after being exposed to 20 bar of syngas for 30 min at 270 °C.

Supplementary Table 5 |Assignment of DRIFTS peaks.

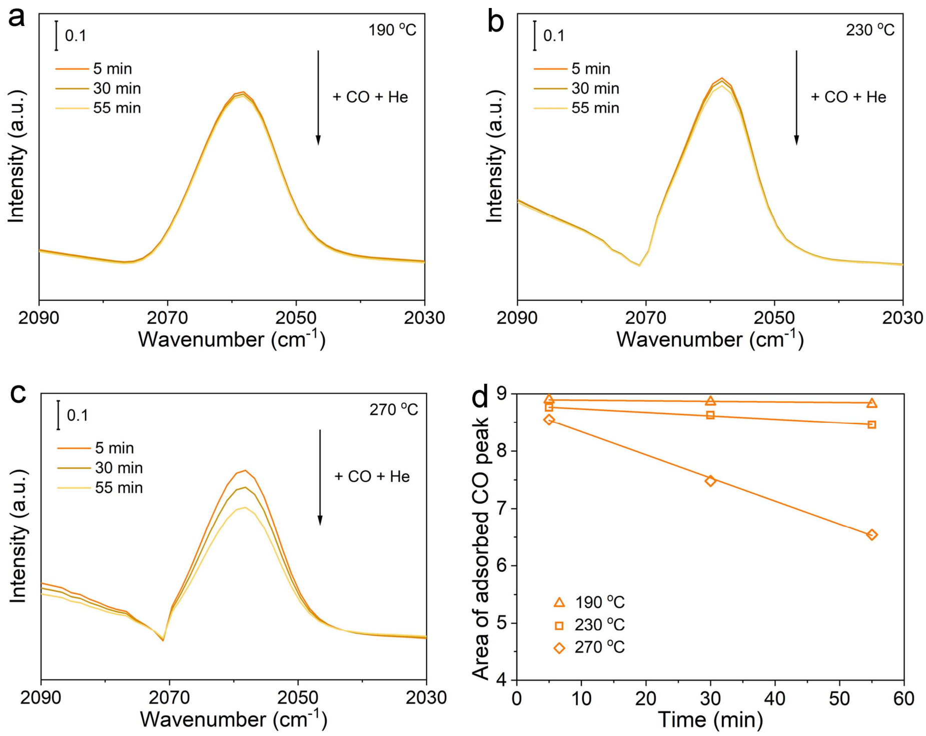
Wavenumber (cm^{-1})	Assignment	References
2360, 2336	Gaseous CO_2	1
2955	Asymmetrical stretching vibration of C-H bonds in CH_3*	2, 3
2919	Asymmetrical stretching vibration of C-H bonds in CH_2*	4
2849	Symmetrical stretching vibration of C-H bonds in CH_2*	5
1741	Stretching vibration of C=O bonds in CHO*	6,7



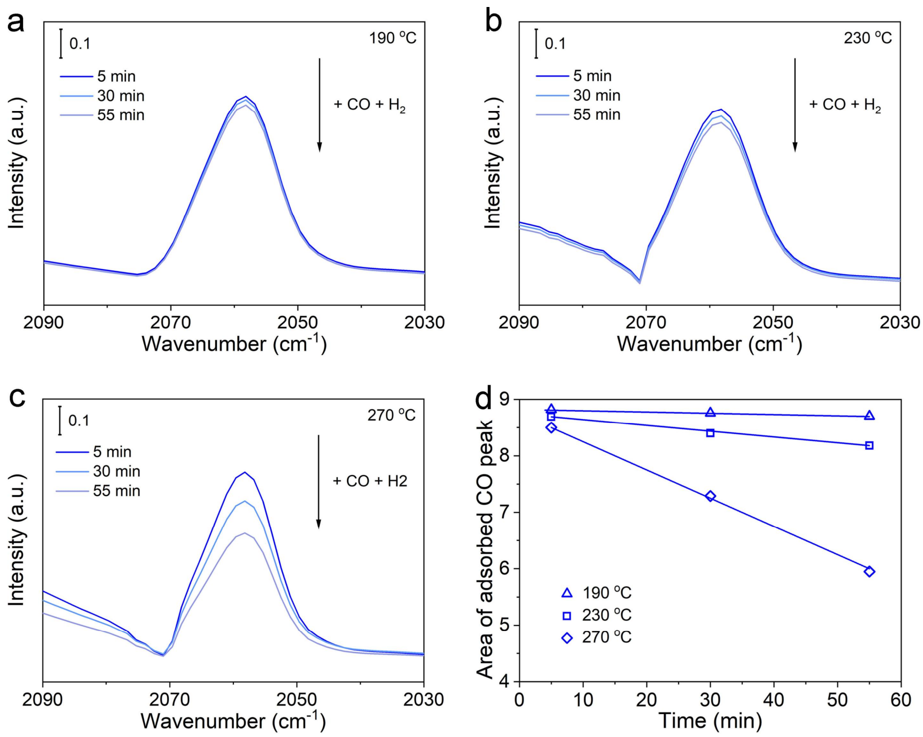
Supplementary Figure 23 | The schematic diagram for different CO dissociation pathways of Fe_3O_4@χ-Fe_5C_2 nanocubes and Fe_3O_4@χ-Fe_5C_2 octahedra.



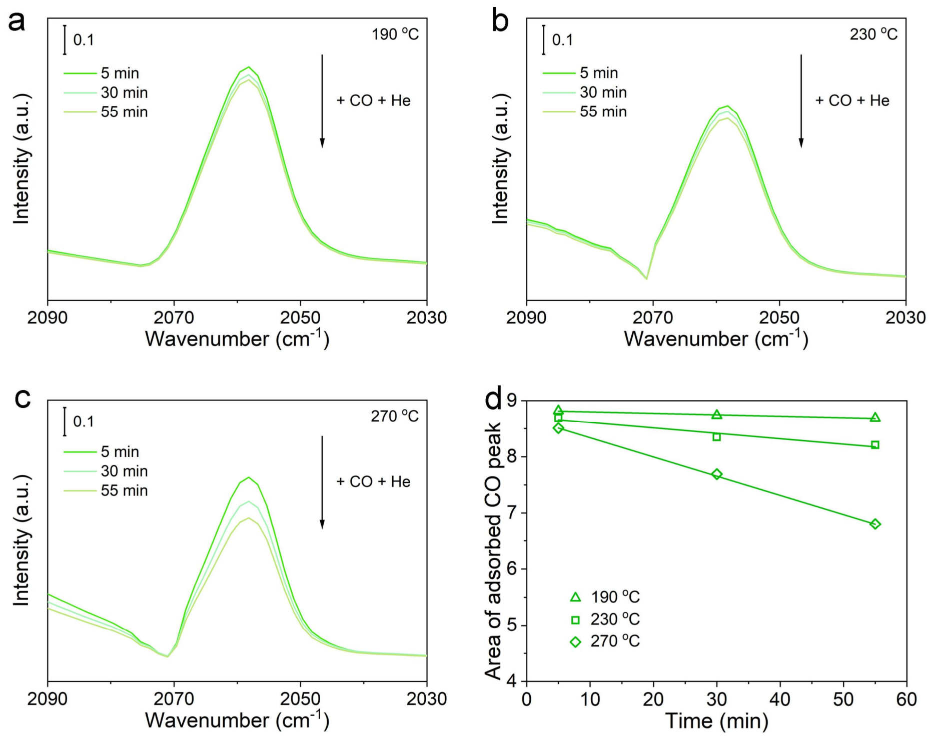
Supplementary Figure 24 | Mechanistic studies of Fe_3O_4@χ-Fe_5C_2 nanocubes for CO dissociation. (**a-c**) *In-situ* DRIFTS spectra of CO adsorption on Fe_3O_4@χ-Fe_5C_2 nanocubes after being exposed to CO and purged with H_2 at (**a**) 190 °C, (**b**) 230 °C, and (**c**) 270 °C. Specifically, Fe_3O_4@χ-Fe_5C_2 nanocubes were exposed to 1 bar of CO for 30 min. Subsequently, the spectra were recorded at 5, 30, and 55 min after being purged H_2. (**d**) Time course of the area of adsorbed CO peak over Fe_3O_4@χ-Fe_5C_2 nanocubes in H_2.



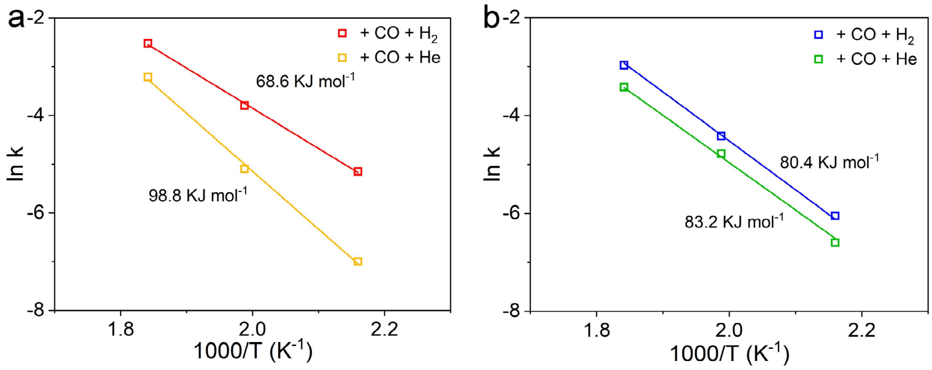
Supplementary Figure 25 | Mechanistic studies of Fe_3O_4@χ-Fe_5C_2 nanocubes for CO dissociation. (**a-c**) *In-situ* DRIFTS spectra of CO adsorption on Fe_3O_4@χ-Fe_5C_2 nanocubes after being exposed to CO and purged with He at (**a**) 190 °C, (**b**) 230 °C, and (**c**) 270 °C. Specifically, Fe_3O_4@χ-Fe_5C_2 nanocubes were exposed to 1 bar of CO for 30 min. Subsequently, the spectra were recorded at 5, 30, and 55 min after being purged He. (**d**) Time course of the area of adsorbed CO peak over Fe_3O_4@χ-Fe_5C_2 nanocubes in He.



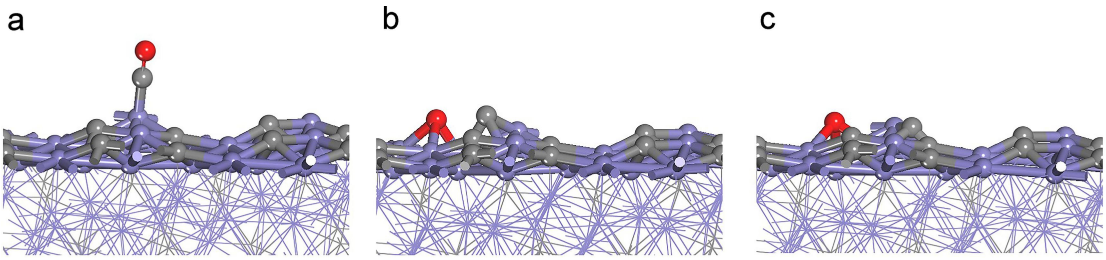
Supplementary Figure 26 | Mechanistic studies of Fe_3O_4@χ-Fe_5C_2 octahedra for CO dissociation. (**a-c**) *In-situ* DRIFTS spectra of CO adsorption on Fe_3O_4@χ-Fe_5C_2 octahedra after being exposed to CO and purged with H_2 at (**a**) 190 °C, (**b**) 230 °C, and (**c**) 270 °C. Specifically, Fe_3O_4@χ-Fe_5C_2 octahedra were exposed to 1 bar of CO for 30 min. Subsequently, the spectra were recorded at 5, 30, and 55 min after being purged H_2. (**d**) Time course of the area of adsorbed CO peak over Fe_3O_4@χ-Fe_5C_2 octahedra in H_2.



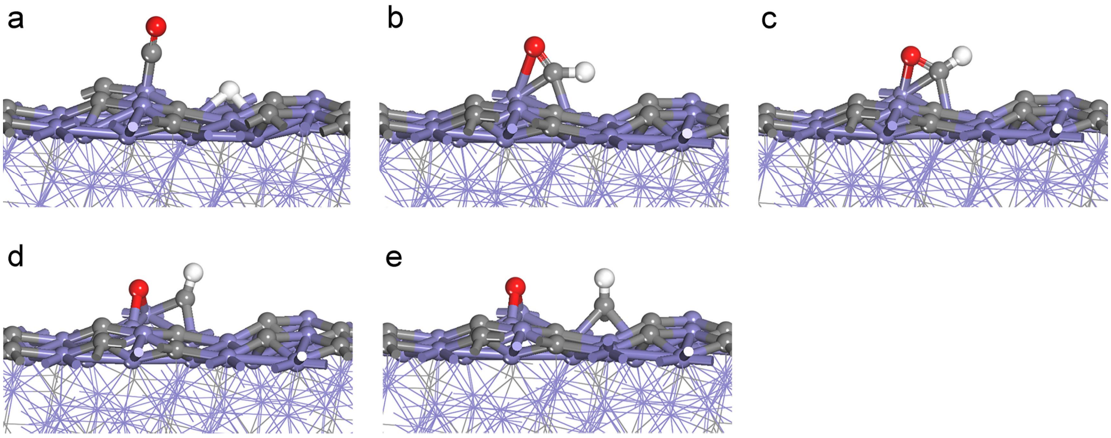
Supplementary Figure 27 | Mechanistic studies of Fe_3O_4@χ-Fe_5C_2 octahedra for CO dissociation. (**a-c**) *In-situ* DRIFTS spectra of CO adsorption on Fe_3O_4@χ-Fe_5C_2 octahedra after being exposed to CO and purged with He at (**a**) 190 °C, (**b**) 230 °C, and (**c**) 270 °C. Specifically, Fe_3O_4@χ-Fe_5C_2 octahedra were exposed to 1 bar of CO for 30 min. Subsequently, the spectra were recorded at 5, 30, and 55 min after being purged He. (**d**) Time course of the area of adsorbed CO peak over Fe_3O_4@χ-Fe_5C_2 octahedra in He.



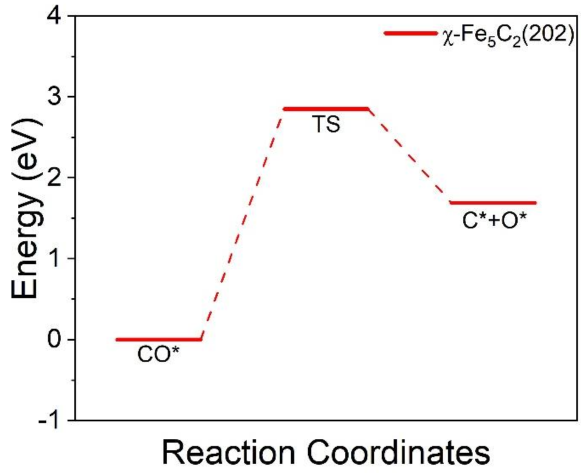
Supplementary Figure 28 | (a, b) Arrhenius plots and activation energy of (**a**) Fe_3O_4@χ-Fe_5C_2 nanocubes and (**b**) Fe_3O_4@χ-Fe_5C_2 octahedra after being exposed to CO and purged with He or H_2.



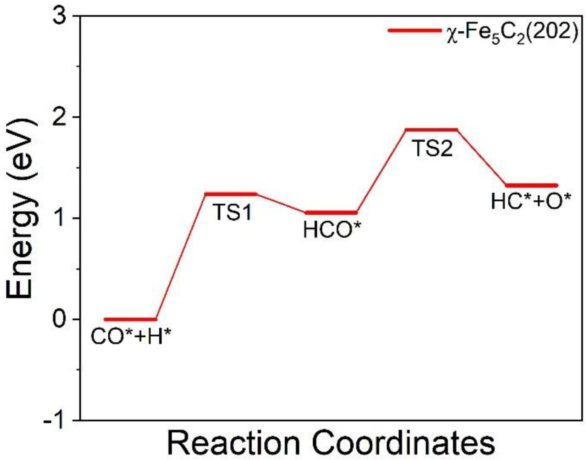
Supplementary Figure 29 | Calculated structure models of the direct CO dissociation route on the χ-Fe_5C_2(202) facet. Models of (**a**) CO*, (**b**) TS, and (**c**) C* + O*.



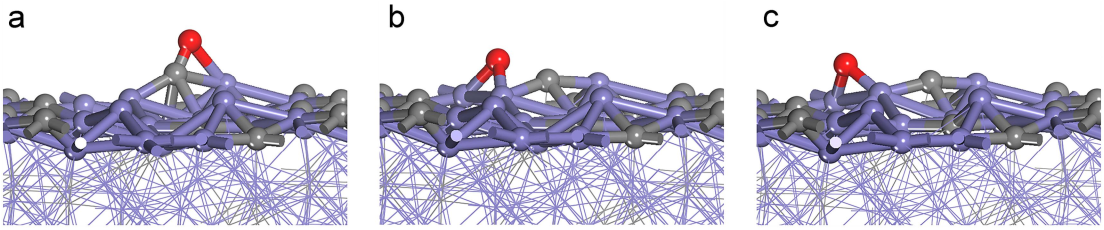
Supplementary Figure 30 | Calculated structure models of the hydrogen-assisted CO dissociation route on the χ-Fe_5C_2(202) facet. Models of (**a**) CO* + H*, (**b**) TS1, (**c**) HCO*, (**d**) TS2, and (**e**) HC* + O*.



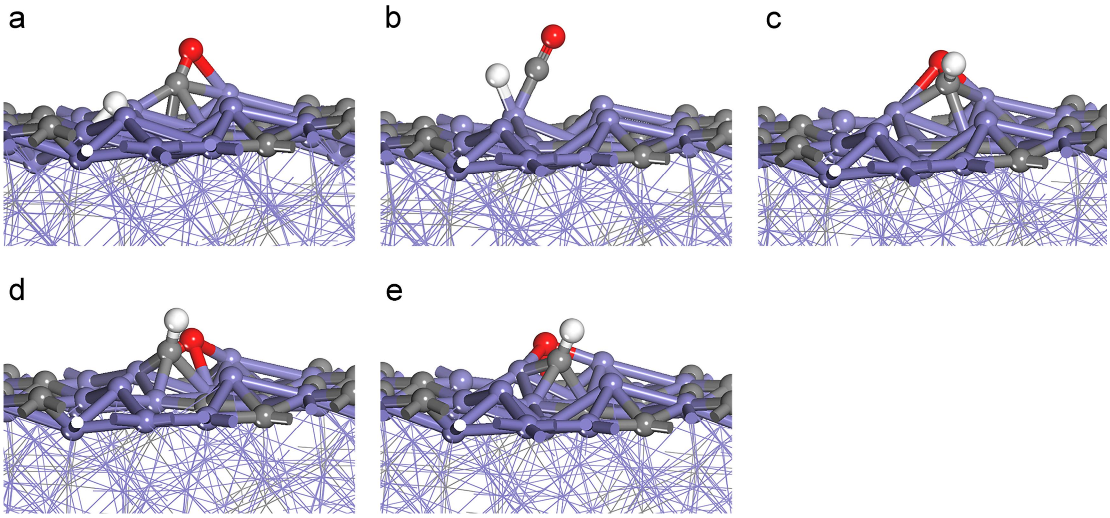
Supplementary Figure 31 | Energy barrier of the direct CO dissociation route on the χ-Fe_5C_2(202) facet.



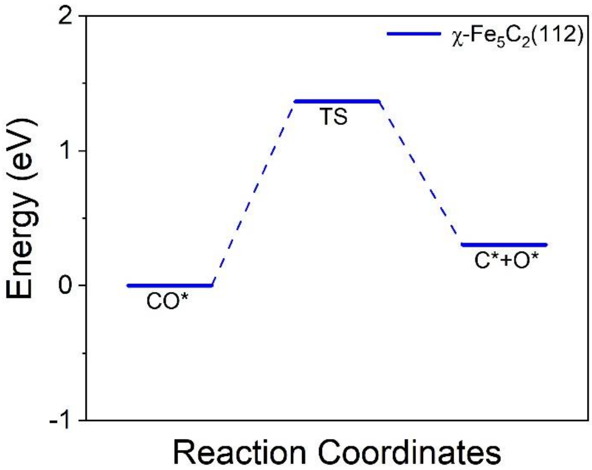
Supplementary Figure 32 | Energy barrier of the hydrogen-assisted CO dissociation route on the χ-Fe_5C_2(202) facet.



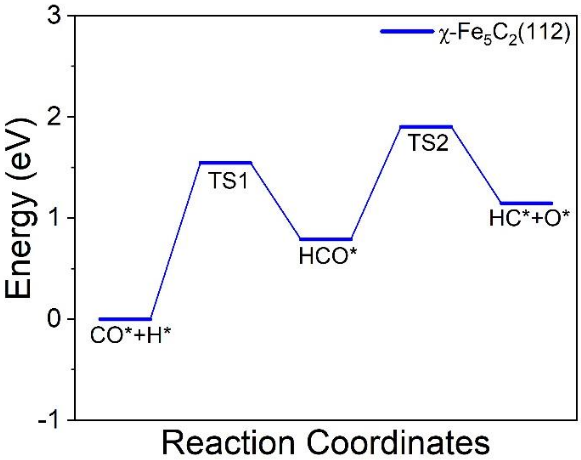
Supplementary Figure 33 | Calculated structure models of the direct CO dissociation route on the χ-Fe_5C_2(112) facet. Models of (**a**) CO*, (**b**) TS, and (**c**) C* + O*.



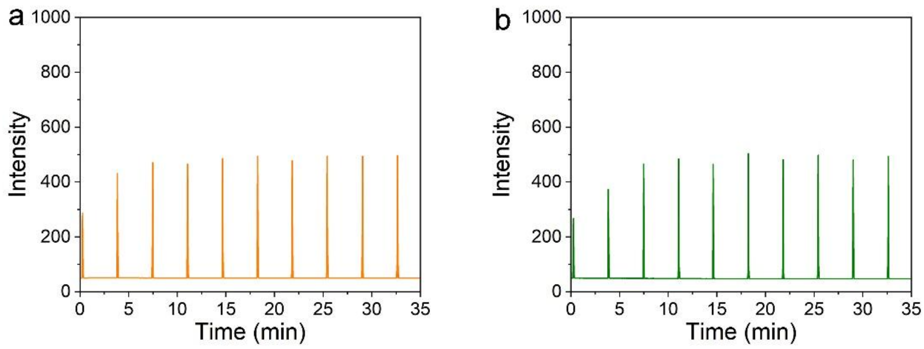
Supplementary Figure 34 | Calculated structure models of the hydrogen-assisted CO dissociation route on the χ-Fe_5C_2(112) facet. Models of (**a**) CO* + H*, (**b**) TS1, (**c**) HCO*, (**d**) TS2, and (**e**) HC* + O*.



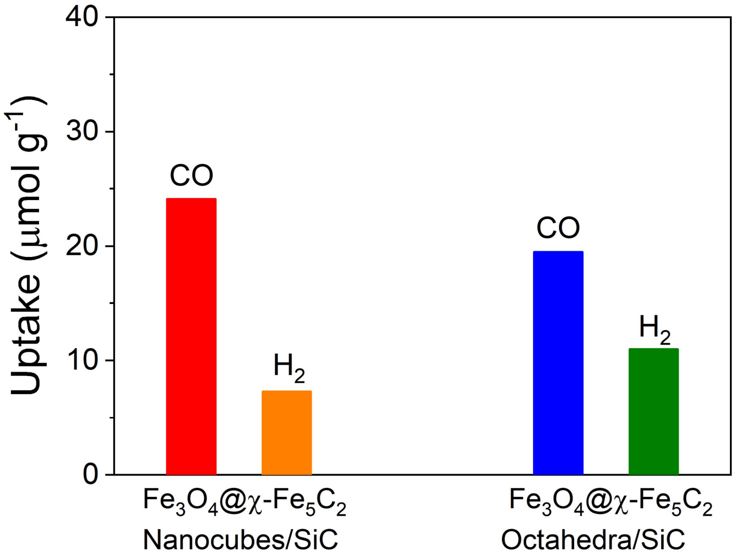
Supplementary Figure 35 | Energy barrier of the direct CO dissociation route on the χ-Fe_5C_2(112) facet.



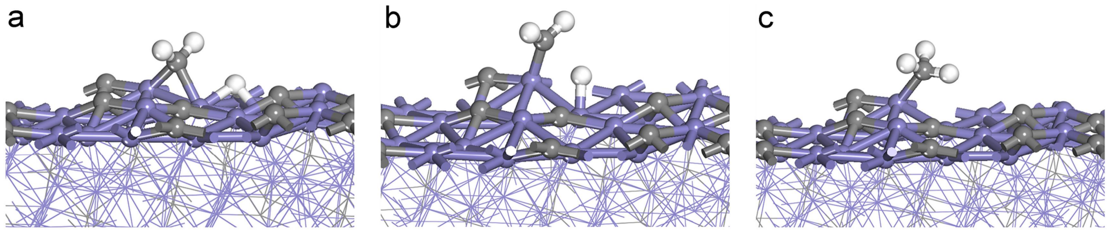
Supplementary Figure 36 | Energy barrier of the hydrogen-assisted CO dissociation route on the χ-Fe_5C_2(112) facet.



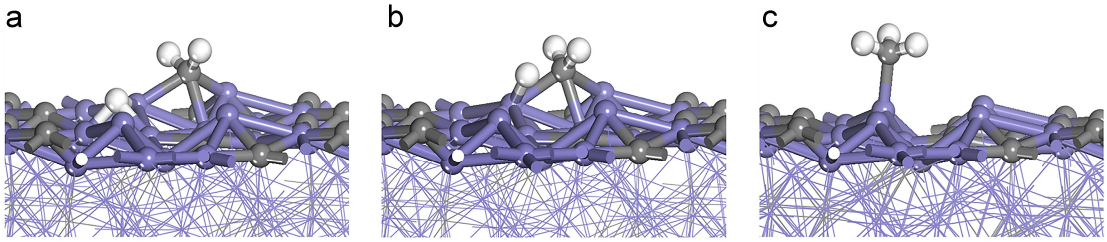
Supplementary Figure 37 | H_2 pulse profiles of (**a**) $Fe_3O_4@\chi\text{-}Fe_5C_2$ nanocubes/SiC and (**b**) $Fe_3O_4@\chi\text{-}Fe_5C_2$ octahedra/SiC.



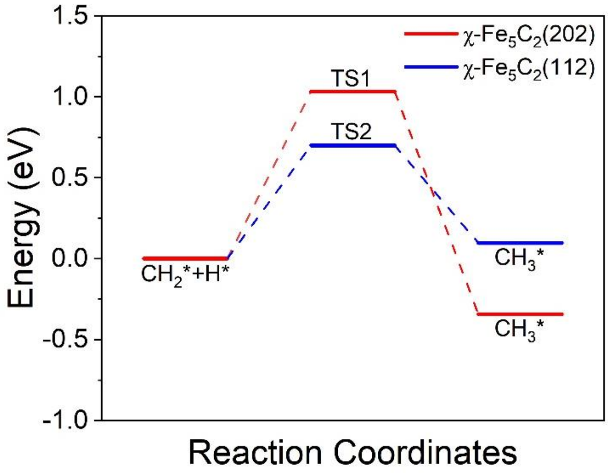
Supplementary Figure 38 | The amount of adsorbed CO and H_2 over Fe_3O_4@χ-Fe_5C_2 nanocubes/SiC and Fe_3O_4@χ-Fe_5C_2 octahedra/SiC.



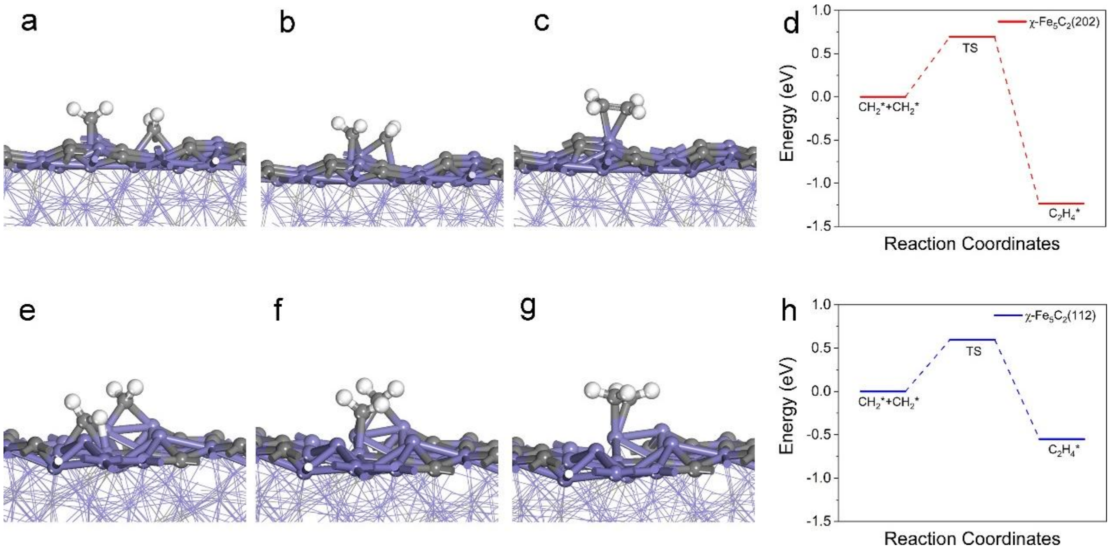
Supplementary Figure 39 | Calculated structure models of CH_2*+H* on the χ-Fe_5C_2(202) facet. Models of (**a**) CH_2*+H*, (**b**) TS1, and (**c**) CH_3*.



Supplementary Figure 40 | Calculated structure models of CH_2* + H* on the χ-Fe_5C_2(112). Models of (**a**) CH_2* + H*, (**b**) TS2, and (**c**) CH_3*.



Supplementary Figure 41 | Comparison in energy barriers of CH_2* + H* over χ-Fe_5C_2(202) and χ-Fe_5C_2(112) facets.



Supplementary Figure 42 | Calculated structure models and energy barriers of CH_2*+CH_2*. Models of (**a**) CH_2*+CH_2*, (**b**) TS, (**c**) C_2H_4* over χ-Fe_5C_2(202). (**d**) Energy barriers of CH_2* + CH_2* over χ-Fe_5C_2(202). Models of (**a**) CH_2*+CH_2*, (**b**) TS, (**c**) C_2H_4* over χ-Fe_5C_2(112). (**d**) Energy barriers of CH_2* + CH_2* over χ-Fe_5C_2(112).

Supplementary References

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